

**Program Design and Evaluation Strategy
for
Opportunity NYC-Family Rewards**

*A Comprehensive
Conditional Cash Transfer (CCT)
Pilot Program for New York City*

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in collaboration with Seedco
for the New York City
Center for Economic Opportunity (CEO)

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Introduction

New York City has launched a major antipoverty initiative called Opportunity NYC—Family Rewards, which is a conditional cash transfer (CCT) program intended to help families break the cycle of intergenerational poverty. The plan has been inspired by successful programs of this kind operating in other countries, including Mexico. The program is being tested as a special demonstration project in six of New York’s highest-poverty communities, with its effectiveness carefully assessed through a randomized control trial.

Opportunity NYC grows out of the work of the Commission on Economic Opportunity (also known as the Poverty Commission), appointed by Mayor Michael Bloomberg in early 2006, to issue recommendations for reducing the number of New Yorkers living in poverty. It is a direct response to the high levels of poverty that persist in some City neighborhoods despite considerable regional economic growth and changes in federal, state, and local welfare and related policies over the last two decades. In 2006, when the Poverty Commission deliberated, the official poverty rate in the City was 19.1 percent. In the communities of concentrated poverty in Manhattan, Brooklyn, and the Bronx that are the focus of this demonstration, poverty rates were near or above 40 percent—a level of poverty that many experts define as “extreme poverty.”

Family Rewards is one of three random assignment demonstration projects initiated in 2007 by City’s Center for Economic Opportunity (CEO) to test the effectiveness of incentive-based poverty reduction strategies. The other two are:

- A new workforce-focused demonstration, called Opportunity NYC—Work Rewards, involving low-income recipients of governmental rent subsidies through New York City’s Housing Choice Voucher (Section 8) programs.
- An education-focused incentives program, called Spark, which is designed to improve school performance of fourth and seventh graders by rewarding performance on a series of standardized tests administered over the course of the academic year.

The three projects differ in important ways, but all three use cash payments to promote activities that will build human capital and improve children’s chances of escaping poverty when they grow up. Together they make up a set of incentives-based demonstrations known collectively as Opportunity NYC, all of which are being rigorously evaluated using randomized control trials. A consortium of private funders is supporting these studies.

CEO guided the process of designing the Family Rewards and Work Rewards demonstrations in collaboration with various City agencies¹ and in partnership with two national,

¹ Staff from the New York City Department of Health and Mental Hygiene, Department of Education, and Human Resources Administration/Department of Social Services, Department of Consumer Affairs, and Department of Small Business Services were planning partners on Family Rewards, while the Department for Housing Preservation and Development and the New York City Housing Authority were the main planning partners for the housing-based demonstration. In addition, six academic experts provided a critical peer review of and helpful feedback on an earlier draft of this paper.

New York-based nonprofit organizations: MDRC, a nonpartisan social policy research organization, and Seedco, a social services intermediary organization. Seedco, in partnership with a small network of local community organizations, is managing the operations for the CCT components, including CCT enrollment services and the CCT verification and payment system. MDRC is conducting the evaluation.

This paper discusses the program model and research design for Family Rewards. It describes the problems the demonstration is attempting to address, the communities in which the program is being tested, the intervention plan, the rationale behind its multiple components, the sample on which it will be tested, and the overall analysis strategy for assessing the project's effectiveness.

Program Overview

Several decades of research have documented the costs of poverty—to children, to their families and communities, and to society as a whole.² Family Rewards is a two-generation initiative to reduce poverty, and it includes both shorter-term and longer-term poverty reduction goals. It is being tested in six community districts in the Bronx, Brooklyn, and Manhattan that encompass a variety of neighborhoods, including the following areas: Central and East Harlem in Manhattan; Brownsville and East New York in Brooklyn; and Morris Heights/Mount Hope and East Tremont/Belmont in the Bronx. The program includes no new social services or case management. Instead, it will attempt to use the offer of a new set of cash transfers in strategic ways to achieve three inter-related objectives:

(1) to lessen immediate income-related hardships for poor families, (2) to encourage poor families to increase—or sustain—positive efforts to improve their own futures, and (3) to help and encourage poor families to invest in their children's futures. Thus, the payments to families are to function both as an *income supplement* that comes in the form of *incentives* to sustain and/or support efforts to improve their situations, and as an inducement to take full advantage of existing institutions and programs that can help them succeed.

The monetary payments will be awarded only when households meet specific conditions in three key areas: children's education, family preventive health care practices, and parents' workforce efforts. For example:

Sponsor

New York City Center for
Economic Opportunity (CEO)

Funders

The Rockefeller Foundation
The Starr Foundation
Bloomberg Philanthropies
American International Group (AIG)
Robin Hood Foundation
Open Society Institute
The Broad Foundation
New York Community Trust
The John D. and Catherine T.
MacArthur Foundation
The Annie E. Casey Foundation
Tiger Foundation

² See, for example, Huston, 1991; Chase-Lansdale and Brooks-Gunn, 1995; Duncan and Brooks-Gunn, 1997; Brooks-Gunn, Duncan, and Aber, 1997a, 1997b; Nilsen, 2007; and Holzer et al., 2007.

- **Education-based conditions** include children's superior attendance in school, sustained high achievement or improved performance on standardized tests, and parental engagement in children's education.
- **Health-based conditions** include maintaining adequate health coverage for all children and adults in participant households as well as age-appropriate preventive medical and dental visits for each family member.
- **Workforce-related conditions** include sustaining full-time work and participation in approved education or job training while working either part time or full time.

Although each of these components targets specific problems that contribute to long-term and intergenerational poverty, the Family Rewards model builds on the recognition that *sustained* achievements in any one of these areas may be aided by progress in the others. For example, children's school progress may benefit from improvements in health care and efforts to catch and address health problems early. Children's health and education may both benefit from growing up in a household that has more economic resources at its disposal, for which parents' sustained employment is critical. At the same time, parents' ability to work steadily and seek new opportunities can be impeded if they or their children are not healthy, and if they struggle with children who are detached from school and performing poorly. Moreover, children's health as adults may be influenced by their education, which can affect their understanding of good health practices and healthy lifestyle choices. A multi-dimensional conditional cash transfer program also makes it easier to construct a transfer that has a sizable total cash value without attaching excessive amounts of money to any one area of progress. For all of these reasons, it is hoped that by combining all three components (education, health, and workforce) into a single, two-generation package will be more powerful than focusing on any one or two of these components alone. They are expected to function in mutually supporting ways to achieve Family Rewards' ambitious goals: alleviating poverty in the short run, improving child and adult health and education outcomes along with parental self-sufficiency in the intermediate term, and reducing intergenerational poverty in the long term.

This initiative targeted families in the selected community districts who had incomes at or below 130 percent of the federal poverty level. This standard is the same as the eligibility standard used for food stamps and a number of other benefit programs that serve very low-income families. In other words, it is a widely accepted benchmark for identifying families most in need of government income transfer programs. To be eligible for Family Rewards, families are not required to participate in any government benefit programs; however, their incomes must be within the 130-percent-of-poverty cutoff level. The program also targeted families who had school-age children living with them. In particular, for reasons explained later in this paper, eligible families had to have at least one child in either the fourth, seventh, or ninth grade, most of whom were identified from student lists provided by the New York City Department of Education.³ However, once a family has enrolled in the program, *all* school-age children in the family are eligible for the program's education and health-focused incentives payments,

³ Families whose children were enrolled in a parochial or other private school but who otherwise met the program's eligibility criteria were also allowed to volunteer for the study.

regardless of their grade level.⁴ Overall, 4,778 families (with 5,051 adults and 11,489 children) are in the study sample, with half assigned to the program group and half assigned to the control group.⁵

Participating families in the program group receive bi-monthly cash payments that can total \$4,000-\$6,000 per year (or more) for three years, as long as they meet the specified conditions across the three domains of activity. Families also received help opening bank accounts (and an added incentive to do so) so that the payments can be transferred electronically and accessed via debit cards. The intervention seeks to raise family income substantially in the short-term – through the payments themselves and by encouraging parents to enter and sustain full-time work – while promoting positive health and educational efforts that will improve children’s future prospects, and investments in parents’ own human capital development.

Seedco has assembled a network of local organizations in the designated community districts to assist in implementing Family Rewards. Called Neighborhood Partner Organizations (NPOs), these organizations recruited and enrolled eligible families into the research sample (explaining random assignment as well), and have provided detailed program orientations for the program group, explaining the program’s incentives offer and the procedures for making claims. Over the life of the program, the NPOs will provide ongoing customer service to participants who request help in making claims, who need additional guidance on program rules and procedures, or who request assistance in finding services that can help them meet the program conditions

2007 HHS Poverty Guidelines		
Persons in Family or Household	48 Contiguous States and D.C.	130 Percent of the Federal Poverty Line
1	\$10,210	\$13,273
2	13,690	17,797
3	17,170	22,321
4	20,650	26,845
5	24,130	31,369
6	27,610	35,893

(e.g., information on where to get homework help for their children, how to find a doctor, where to get help finding jobs and training). NPOs will also conduct workshops to provide program clarifications as well as information and support for completing incentivized activities. In addition to this ongoing NPO support, participants will also be able to get assistance through a telephone helpline and Web site operated by Seedco. Thus, although the program includes no pro-active case management with routinely scheduled appointments for all participants, families will have several avenues through which they can get information on services and procedures.

The program design for Family Rewards calls for the incentives to be offered to each participating family for three years. However, whether a third year is actually included is depends on the availability of funds.

⁴ Family members eligible for payments included children 18 years old or younger and their custodial parents or legal guardians. A custodial parent’s cohabiting spouse or legally registered domestic partner who was not the child’s own parent was also eligible.

⁵ This number does not include a very small number of families who were randomly assigned but subsequently withdrew from the study.

The Study Neighborhoods

Table 1 lists the six community districts (CDs) selected for testing Family Rewards, and notes the local neighborhoods they largely encompass. In the Bronx, these neighborhoods include East Tremont, Bathgate, Belmont, West Farms; and Morris Heights, University Heights, Fordham, and Mount Hope in the Mid- and Northwest Bronx. In Brooklyn, they include Ocean Hill-Brownsville and several East and South-East Brooklyn communities, including East New York, New Lots, and the large subsidized affordable housing development of Starrett City. In Manhattan, they include Central Harlem and East Harlem.

As suggested by Table 2, these districts are predominantly black and Latino. On average across the districts, about 50 percent of residents are black and 41 percent are Latino—compared with 25 percent and 27 percent, respectively, for New York City as a whole. At the same time, many neighborhoods within these districts have distinct racial and ethnic identities. The Bronx neighborhoods are predominantly Latino, as is East Harlem. However, East Harlem is one of the City’s traditional seats of Puerto Rican culture, while Community District 5 in the Bronx has a large Dominican population. Central Harlem and Ocean Hill-Brownsville in Brooklyn are predominantly African-American⁶.

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Table 1

Key Neighborhoods in Selected Community Districts

Community District	Key Neighborhoods
Bronx	
Community District 5	Morris Heights, University Heights, Fordham, Mount Hope
Community District 6	East Tremont, Bathgate, Belmont, West Farms
Brooklyn	
Community District 5	East New York, New Lots, Starrett City
Community District 16	Brownsville, Ocean Hill
Manhattan	
Community District 10	Central Harlem
Community District 11	East Harlem

On the whole, these neighborhoods have fewer numbers of immigrant families than the city as a whole, with Community District 5 in the Bronx, and Starrett City, New Lots and East New York being home to significantly more households born outside of the US. The proportion

⁶ Community District 5 in Eastern Brooklyn encompasses three distinct neighborhoods, including the very large mixed-income development of Starrett City.

of residents over 25 without high school diplomas among districts averaged 43 percent, compared with 28 percent in the city as a whole. In many neighborhoods – such as Community District 5 in the Bronx and Community District 5 in Brooklyn, the proportion of married families with children is similar to that of the city as a whole. But in all of them, the proportion of single female heads of household with at least one child under 18 is significantly higher than the citywide average.

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Table 2

Key Demographic Information, by Community District

Characteristic	Bronx		Brooklyn		Manhattan		Citywide
	CD 5	CD 6	CD 5	CD 16	CD 10	CD 11	
Total population	128,313	75,688	173,198	85,343	107,109	117,743	8,008,278
Race/ethnicity (%)							
Asian	2	1	4	1	1	3	10
Black/African-American	32	26	49	78	77	36	25
Hispanic/Latino	62	61	38	18	17	52	27
White non-Hispanic/Latino	2	9	5	1	2	7	35
Foreign-born (%)	35	23	33	21	18	21	36
Born in Dominican Republic (%)	19	8	8	2	4	4	5
Born in Ecuador (%)	2	1	1	0	0	1	1
Born in Guyana (%)	1	0	5	3	0	0	2
Born in Jamaica (%)	2	1	3	4	1	0	2
Born in Mexico (%)	2	3	1	0	1	7	2
Not proficient in English (%)	34	31	20	9	11	27	24
No High School Diploma (%) ^a	49	50	41	40	36	44	28
High School graduate or equivalent only (%) ^b	23	24	29	32	27	23	24
Some college (%)	21	18	20	21	23	19	20
College graduate (BA or higher) (%)	8	8	9	7	15	15	27
Total number of households	40,220	24,618	55,042	28,305	45,734	43,318	3,021,588
Married couple with at least one related child under 18 (%)	18	15	20	13	9	12	19
Single female head of household with at least one related child under 18 (%)	32	30	27	34	22	23	13

SOURCE: 2000 Census

^a Education statistics include only 25+ population.

The six community districts were chosen for testing Family Rewards because they are among New York's most persistently disadvantaged. Table 3 illustrates their high rates of poverty and high unemployment, even when economic conditions in the City as a whole are good. For example, in 2000, about 40 percent of the households in these districts had incomes below the poverty level, compared with a citywide rate of 21 percent. The unemployment rate

across the districts was 19 percent, on average, compared with 5 percent for the City as a whole. Considerably higher proportions of residents of these communities, compared with the City's population as a whole, relied on public benefits, including TANF, food stamps, and Medicaid.

Table 4 shows that while these neighborhoods experience many problems that often accompany poverty, they differ on some important dimensions, including housing burden and crime. All suffered from higher asthma hospitalization rates, the proportion of unemployed youth who were also not in school, and in the overall rate of felonies reported per 1000 persons. The six districts as a whole constituted nearly 14 percent of juvenile felony arrests citywide in 2001. However, the number of juvenile felony arrests in these districts varied from a low of 93 in Community District 6 in the Bronx to a high of 211 in Eastern Brooklyn. In Ocean-Hill Brownsville, the number of felonies per 1,000 residents was nearly twice that of neighboring communities represented in district 5 in Eastern Brooklyn. Reflecting some variation in housing

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Table 3

Labor Force, Income, and Benefit Receipt Information, by Community District

Characteristic	Bronx		Brooklyn		Manhattan		Citywide
	CD 5	CD 6	CD 5	CD 16	CD 10	CD 11	
Poverty-level income (%)	41	46	33	43	37	37	21
Of population 16 years and over:							
In the labor force (%)	51	46	51	49	51	47	58
In the labor force and employed (%)	40	37	43	38	41	39	52
Unemployment rate	20	21	16	23	18	17	10
Not in the labor force (%)	50	54	49	51	49	53	42
Females 16 years and over with own children under 18 years in labor force (%)	51	51	56	58	61	51	58
Receiving Public Assistance (%) ^a	17	19	11	16	10	11	5
TANF	8	9	5	7	4	5	2
Safety Net Converted	5	6	3	5	2	3	1
Receiving SSI (%)	9	10	8	9	8	10	5
Receiving Medicaid only (%) ^b	33	27	29	25	25	26	23
Total Medicaid enrollees (%)	58	56	48	50	44	48	33
Total Food Stamp recipients (%)	32	33	25	30	23	26	14

SOURCES: Poverty: New York City Department of Planning (2000 Census). Benefit information: HRA (as of 10/05)

^a Public Assistance is defined as TANF, Safety Net Assistance, and Safety Net Converted.

^b Medicaid program data include Child Health Plus Part A and Family Health Plus.

conditions and problems, the proportion of households with extreme rent burdens (those paying more than 50 percent of income on rent) was nearly 35 percent in the Bronx neighborhoods, but approximately 20 percent in Brooklyn neighborhoods. The Mid and Northwest Bronx neighborhoods, together with East Harlem, had some of the highest proportion of overcrowded rental housing in the city, at 21 percent and 24 percent in Bronx Community District 5 and Manhattan Community District 11 respectively; while overcrowding was much less of a problem in both Brooklyn districts and in Harlem.

Harder to measure, but critical to both the functioning of the initiative and its assessment, is the social service infrastructure of these neighborhoods. While many community and citywide social services providers operate in all these districts, the density and connectedness of providers varies across the different neighborhoods. For example, many Bronx neighborhoods have housing and social service networks that formed around Latino political and activist institutions after rapid demographic change during and after the 1970s. At one extreme of connectedness, Harlem has a dense and interconnected provider network, facilitated by such strong organizations as Harlem Children’s Zone and historic ties to African-American political and cultural institutions. Other neighborhoods, while having many competent agencies, may lack such a deep community-based, inter-connected infrastructure. At the other extreme, East New York has sometimes been described as a neighborhood that struggles to maintain social services after extreme disinvestment and high crime rates impacted the area since the 1970s.

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Table 4

Key Social Indicators, by Community District

Characteristic	Bronx		Brooklyn		Manhattan		Citywide
	5	6	5	16	10	11	
Households that spent one-half or more income on rent (%)	35	32	19	20	23	24	24
Overcrowded rental housing (%)	21	9	8	9	8	24	11
Bank branches in Community District	4	4	10	2	9	8	1,196
Reported felonies per 1000 residents	34	37	20	43	36	32	27
Youths 16-19 not in school and not in labor force (%)	10	8	6	7	7	8	5
Youths 16-19 not in school and not high school graduates (%)	17	14	17	19	8	12	11
Number of juvenile felony arrests (under 16 years)	114	93	211	178	126	129	6,054
Births to teen mothers (%)	16	16	15	17	14	18	9
Infant mortality rate	6.8	6.9	5.6	12.7	13.2	10.0	5.6
Low birthweight babies (%)	9	9	11	15	12	10	8
Asthma hospitalization rate	6.2	8.8	7.2	9.1	7.1	10.5	4.7
Pneumonia hospitalization rate	2.1	2.3	3.1	4.3	2.0	3.0	2.1
Child health clinics in district	0	1	1	2	3	1	38
Voluntary or HHC hospitals in district	0	1	0	0	1	3	79

SOURCE: Citizen's Committee for Children website

Lessons from Past Incentives Programs

Family Rewards may be the first large-scale program to combine financial incentives in the health, education and workforce domains, but many earlier programs—both in the U.S. and abroad—have used incentives to promote specific activities in one or two of these areas. Several of these initiatives have been rigorously evaluated.

This section briefly discusses some of the key evaluation results available to date. The section first discusses “comprehensive” conditional cash transfer models that target more than one domain of activity. The discussion then turns to programs that focus on a single area (i.e., education or health or work). Although many of the examples discussed below come from abroad, several of the programs originated in the U.S. welfare system, which has increasingly taken on some of the characteristics of a CCT in recent years.

In considering these various models, it is important to draw a distinction between programs that used incentives to promote a particular kind of action (e.g., attending school) and programs—like Family Rewards—that have the dual goal of promoting specific actions *and* substantially raising participants’ income. Thus, some of the programs discussed below had two kinds of effects—they triggered impacts on the targeted behaviors and, by raising family income, generated additional indirect effects on measures of family or child well-being that are linked to poverty.

Comprehensive CCT models

Mexico’s *Oportunidades* program (originally called Progresá) has been described as the inspiration for Family Rewards. Launched in 1997, this program now serves 25 million very low-income Mexicans—about one-fourth of the total population.⁷ Its payments are designed to promote preventive health activities, nutrition, and school attendance among children. Several other developing countries—including Brazil, Chile, Colombia, Honduras, and Nicaragua—have created conditional cash transfer programs modeled at least in part on *Oportunidades*. Most of these programs use payments to promote health/nutrition and education behaviors. However, like *Oportunidades* (and unlike Family Rewards), most of them also include some supply-side interventions such as additional health services. This is especially important in poorer countries where the existing supply of services is very limited. Several of the CCT initiatives have been rigorously evaluated, often by randomly selecting communities to phase in the program earlier (the treatment group) or later (the control group). A full review of this research is beyond the scope of this paper, but some results are briefly reviewed below.

Several of the CCT programs have generated positive impacts on school enrollment rates. For example, the Mexican program raised enrollment rates at the secondary school level by 7-9 percentage points for girls (from a baseline enrollment rate of 67 percent) and by 3-6 percentage points for boys (from a baseline level of 73 percent). The program in Colombia also raised enrollment at the secondary level, by 4-6 percentage points in rural areas and 12-14 percentage points in urban areas (where the baseline enrollment rate was 64 percent). In Nicaragua, baseline enrollment at the primary school level was only 69 percent, and the program produced a 22-percentage point increase. Impacts on school attendance have been more mixed. In Nicaragua, impacts on attendance were larger than impacts on enrollment, but in Mexico, the reverse was true.⁸ Over a follow-up period of more than five years, the Mexican program generated impacts on grades of schooling attained, but generally did not affect scores on

⁷ For a comprehensive review of the program’s design, implementation, and effects, see Levy, 2006.

⁸ Rawlings 2004.

achievement tests.⁹ Not surprisingly, many of the programs also produced decreases in child labor.

Several of the programs have also improved child health and nutrition. Oportunidades reduced the prevalence of stunting, increased food consumption, and raised immunization rates. Nicaragua's program also reduced stunting and resulted in improved diets. On the other hand, a program in Honduras had no impact on stunting or nutrition.¹⁰

In the U.S., at least two small-scale programs have used financial incentives to encourage a range of positive activities. Neither has been rigorously evaluated:

- The Family Independence Initiative provides monetary rewards for actions like improving children's grades, improving credit scores, or enrolling in a health insurance program. Families can receive up to \$2,000 annually, plus an additional \$4,000 as part of a matching savings program. An initial pilot in Oakland served 140 families, and program sponsors reported that household incomes rose 26 percent in 18 months. Small pilots have also operated in Hawaii and San Francisco.
- A Chicago program, Pathways to Rewards, allows participants to earn points for an individualized set of goals such as holding a job, going to scheduled medical appointments, improving their children's school attendance, and paying the rent on time. The points can be redeemed for gift certificates or checks made out to particular stores or vendors (e.g., a utility company). As of early 2008, 167 families participated in the program, 450 individuals (adults and children) set goals, and 81 percent of them have met one or more goals.¹¹

Single-domain models

A number of programs in the U.S. and elsewhere have used financial incentives to target one or another of the domains included in Family Rewards. Although narrower in scope than the comprehensive models discussed in the previous section, some of these models also had broad poverty-reduction goals and transferred substantial amounts of income.

- **Education**

In the 1990s, several U.S. states introduced school attendance requirements for recipients of cash welfare (originally AFDC and later TANF). Most of these "learnfare" models penalized recipients whose children did not attend school regularly (one pilot program in Florida also required parents to attend parent-teacher conferences), but at least two of them—one in California and one in Ohio—combined financial bonuses and penalties. Both of these programs targeted teenage custodial parents receiving welfare and both were evaluated using random assignment designs:

⁹ Berhrman, Parker, and Todd 2006.

¹⁰ Sridhar and Duffield 2006.

¹¹ Personal communication with Suzanne Wagner, Project Match.

- Ohio's statewide Learning, Earning, and Parenting (LEAP) program increased the monthly welfare grant by \$62 if the teen attended school (or a GED program) regularly and reduced the grant by \$62 if she did not. Case managers monitored teens' attendance and provided assistance as needed. The program led to some increases in school enrollment and attendance, but had few impacts on school completion, except for a modest increase in GED receipt among teens who were enrolled in school when they started the program.¹²
- The Cal-Learn program provided \$100 bonuses up to four times a year based on the student's report card grades. Sanctions of \$100 were imposed for missing or "inadequate" report cards. In addition, teens received a \$500 bonus for graduating from high school or obtaining a GED. The evaluation used a 4-group design to compare "full Cal-Learn" (including both financial incentives and case management), financial incentives only, case management only, and a no-treatment control group. Like LEAP, full Cal-Learn generated a significant increase in GED receipt but had no impact on high school graduation. Full Cal-Learn was more effective than either of the single-component treatments.¹³

Two experiments in Israel provided generous financial incentives to students who passed a series of national high school exams that are required by universities and some jobs. The first experiment, using individual random assignment within schools, did not find impacts, but the second, which randomly assigned schools, did. The authors speculated that "school wide mobilization" may have been played a role in generating the impacts.¹⁴

An incentive program implemented in schools serving underprivileged students in Texas offered financial incentives to both students and teachers for passing grades on advanced placement (AP) examinations. Using a difference-in-differences methodology, an evaluation found that the AP incentive program was associated not only with an increase in AP test-taking and scores, but also with increases in the number of students scoring above 1100 on the SAT or 24 on the ACT and in the number of students who matriculate in college in Texas. Drawing on anecdotal evidence, the analysis suggests that increased AP participation in schools that implemented the AP incentives program was driven not simply by efforts of teachers and students to maximize their rewards, but also by better access to AP courses, changing norms among teachers, guidance counselors, and students that became more supportive of taking AP courses, and better student information about AP courses.¹⁵

The UK's Education Maintenance Allowance (EMA) provides financial incentives to low-income 16-18 year olds to encourage youth to stay in school beyond the age of compulsory schooling. Several variants of the model have been tried, all of which include weekly payments of £30-£40 per week for attendance, plus larger retention and achievement bonuses. In some variants, payments are issued to parents and in others directly to youth. An evaluation compared youth in EMA pilot sites with youth in comparison areas and found positive impacts

¹² Bos and Fellerath, 1997.

¹³ Mauldon et al., 2000.

¹⁴ Angrist and Lavy, 2002.

¹⁵ Jackson, 2007.

on education participation, particularly for young men in urban areas and those from lower socio-economic subgroups.¹⁶ The program has since been rolled out nationally, with payments issued directly to students' bank accounts.

Finally, the Quantum Opportunity Program (QOP), a demonstration program targeted to at-risk ninth graders, used financial incentives to promote participation in program activities, but in conjunction with a rich set of services. For example, participants could receive \$1 to \$1.33 per hour of participation in QOP activities, and an equivalent amount was deposited into an accrual account payable after the youth finished high school and enrolled in college. A small pilot produced some encouraging results in a random assignment evaluation, but a larger replication, also evaluated using random assignment, found no impacts on high school graduation, grades, achievement, or risky behaviors.¹⁷

- **Health**

A 2004 paper reviewed 47 studies that tested the use of financial incentives to promote preventive health care activities. About half of the studies were classified as “simple” prevention, targeted to a specific behavior such as screening, immunization, well-child visits, or attendance at an educational session. Most of these programs were aimed at high-risk, low socioeconomic status (SES) populations. Examples include: two \$15 payments for low-income people to attend sex-education sessions on the prevention of transmitted diseases, bus passes for low SES women returning for an abnormal pap smear, and \$20 in coupons for farmer's market fresh produce for low SES women to improve nutrition. The other half of the studies addressed “complex” preventive care such as weight loss, nutrition, or smoking cessation.

The review included random assignment studies, time series, and prospective quasi-experimental designs. Most of the “simple” interventions generated statistically significant impacts and the review concluded that “in the short run, consumer economic incentives are effective for simple preventive care and distinct behavioral goals that are well defined. There isn't sufficient evidence...to say that economic incentives are effective for promoting the long-term lifestyle changes required for health promotion.”¹⁸

An earlier meta-analysis reviewed 11 randomized trials that used financial incentives to enhance patient compliance with medication and medical appointments. All 11 were conducted in the U.S. Four of the programs encouraged parents to obtain medical care for their children (e.g., immunizations, dental care), another sought to induce teen parents to attend postpartum appointments, another promoted compliance with a weight loss program, and several aimed to promote use of prescribed medication. Ten of the 11 studies found that financial incentives promoted better compliance (Giuffrida and Torgerson, 1997).

Currently, a few states are experimenting with incentives to promote health behaviors among Medicaid recipients. The program in West Virginia, which has attracted the most attention and controversy, will scale back basic benefits for most children and parents covered by

¹⁶ Battistin et al., 2004.

¹⁷ Schirm and Rodriguez-Planas, 2004.

¹⁸ Kane, Johnson, Town, and Butler, 2004.

Medicaid, while giving these families access to enhanced benefits if they sign and conform to an agreement with the state. Beneficiaries who do not follow the agreement—for example, do not show up for scheduled appointments, take medication as directed, or receive health screening exams—will revert back to the basic plan. Those who follow the agreement will maintain enhanced benefits, and also receive “credits” to be used for purchasing services that are not covered by Medicaid. Phase-in of the plan began in three rural counties in 2006.¹⁹

- **Work**

In the U.S., financial incentives have long been used to promote work among low-income populations. Many of these programs have aimed to make work more financially attractive relative to welfare, especially because the wages available to welfare recipients are typically low. Some programs have been designed specifically to promote full-time work.

Nonexperimental studies have estimated that the Earned Income Tax Credit, which has the dual goals of raising family income and promoting work, was responsible for a significant proportion of the increase in labor force participation among single mothers in the late 1980s and the 1990s.²⁰

Most states now used earned income disregards to promote work among recipients of Temporary Assistance for Needy Families (TANF) cash assistance (or “welfare”). These policies disregard a portion of earned income in calculating the welfare grant, allowing some working recipients to retain a portion of the grant as an earnings supplement. A random assignment evaluation in Minnesota found that the combination of an enhanced disregard, a work requirement, and welfare-to-work case management generated significant increases in employment and income for long-term welfare recipients. In a typical quarter in the first two years of the study period, the employment rate for long-term recipients in the program group was 50 percent, compared to 37 percent for the control group. The combination also generated improvements in school performance for the elementary-school aged children of long-term recipients (measured both by parent self-reports and by test scores). These improvements may be attributable to the improved economic circumstances of the families (or, perhaps to increased use of center-based child care). The earnings disregard alone, without the work requirement, generated much smaller increases in employment, mostly in part-time work, and little in the way of earnings gains.²¹

Three random assignment demonstration projects—one in the U.S., one in Canada, and one in the United Kingdom—have tested work incentives delivered separately from welfare payments or outside of the welfare system:

- The New Hope Demonstration, which operated in Milwaukee in the 1990s, offered earnings supplements, health insurance, and child care subsidies to low-income people who worked full-time. The program increased stable employment and the likelihood of earnings. It also reduced poverty and improved academic achievement

¹⁹ Steinbrook 2006; Solomon, 2006.

²⁰ Meyer and Rosenbaum, 2001.

²¹ Knox, Miller, and Gennetian 2000; Gennetian, Miller, and Smith 2005.

for boys through the fifth year of the study. For example, in Year 1, the employment rates were 90 percent for the program group and 82 percent for the control group.²²

- The Canadian Self-Sufficiency Project (SSP) offered generous earnings supplements to long-term welfare recipients who left the rolls and worked full time. SSP generated substantial increases in full-time work, reduced poverty, and resulted in improved school performance for young children.²³
- The United Kingdom's Employment Retention and Advancement demonstration (UK ERA), which is still underway, offers a financial bonus lasting two years for sustaining full-time work and a stipend for completing approved training courses while employed. These incentives are offered in combination with post-employment job coaching. Early findings from the evaluation covering the first two years of follow up show substantial positive impacts on the earnings of single parents who began the program while on welfare, and substantial increases in combining work and training among lone parents who entered the program already working part-time.²⁴

It is worth noting that review and synthesis of random assignment studies of 16 welfare employment programs (including the Minnesota program, the Canadian SSP program, and the New Hope demonstration) found that although these programs led to improved outcomes for younger children, they also produced somewhat worse school performance and higher rates of grade repetition for adolescent children of participants.²⁵

• **Implications**

The results cited above show that there is a growing body of experimental literature to suggest that, under certain circumstances, financial incentives in each of the areas targeted by Family Rewards can generate small-to-moderate improvements in health, education, and workforce outcomes. Although these results provide some reason for optimism, it is crucial to note that the Family Rewards demonstration differs from earlier tests in two key respects. First, the comprehensive CCT models were tested in developing countries, obviously a very different environment than New York City. Second, none of the results discussed above were for models that combined incentives in all three domains (education, health, and workforce).

In this regard, the New York City model builds equally on the international and U.S. research on conditional cash transfer programs. It recognizes that Oportunidades was successful in improving school attendance and infant growth and physical development (reducing stunted growth), but did not improve school achievement as measured by standardized tests (perhaps, in part, because of factors related to school quality and the absence of achievement-based incentives). In addition, Oportunidades included no strategy to deal with sustained family self-sufficiency, with the result that the loss of family income was substantial and immediate when a child aged out of school and the family's time-limited eligibility for the program ended. It is

²² Huston et al., 2003.

²³ Michalopoulos et al., 2003.

²⁴ Dorsett et al., 2007; Riccio, et al., 2008.

²⁵ Gennetian et al., 2002.

noteworthy that experts associated with conditional cash transfer programs internationally have begun to discuss the importance of addressing workforce issues as the CCT approach continues to evolve.

The two-generational focus of Family Rewards recognizes that family poverty affects children's school performance and health, both in the immediate term and in the longer term. Thus, a central question for the initiative is whether a model that tackles all three domains (education, health, and workforce) at once, with an explicit two-generational focus, could have cumulative and sustained effects on children and families that are much larger than those seen in earlier, single-domain interventions and that focused just on parents. Also important is whether a model that includes a workforce component can produce improvements in children's school achievement (e.g., performance on achievement tests) that are larger than those found in international efforts that excluded a workforce component.

Theoretical Framework for Family Rewards

At the most fundamental level, the intent behind Family Rewards, as is true of other CCT programs, is to transfer cash to poor families to help alleviate their immediate poverty-related hardships, but to do so in a way that, simultaneously, helps them invest in their own futures. Thus, it seeks to reduce the likelihood (or severity) of persistent poverty *while children are growing up*, and of *second-generation* poverty. If the program “works,” families will have more economic resources in the short-term and enjoy more economic security in the longer run. The children being raised in poor families today will spend less time growing up poor, and they will have a better chance to avoid being poor when they become adults.

Of course, the program is not just about money; in the end, it is about improving families' quality of life. In that regard, it is hypothesized that the program's focus on education, preventive health care, and employment will also lead, in addition to poverty reduction, to a variety of positive outcomes that are important for families but that cannot be monetarized.

The idea that increasing family income among poor families will benefit the well-being of their children over time is supported by a growing body of evidence showing that poverty in childhood is clearly associated with a broad range of educational, health, and social-emotional problems that, in turn, seriously constrain children's future life chances.²⁶ For years, it was a matter of debate whether income poverty itself or the other numerous risk factors associated with poverty had a true causal impact on children's health, education, development, and future life chances.²⁷ Over the last few years, a scientific agreement has emerged that very low income, material hardship, and financial strain have causal influence on children's life trajectories. (The data supporting the claim of causal influences come from the convergence of better longitudinal studies²⁸ and from natural and policy experiments that have effectively raised the income of poor families and evaluated the impact of these increases on children.²⁹)

²⁶ Gershoff, Aber, and Raver, 2003.

²⁷ Duncan and Brooks-Gunn, 1997; Mayer, 1997.

²⁸ Dahl and Lochner, 2005; Duncan and Brooks-Gunn, 1997; Gershoff, Aber and Raver, 2003; Mayer, 2002; McLoyd, 1998; Seccombe, 2000)

²⁹ Costello et al., 2003; Morris and Gennetian, 2003

For these reasons, it is expected that if Family Rewards significantly increases the income of poor families, it will have positive *indirect* effects on young children's education, health, and development, over and above the positive *direct* effects that might result from meeting the education and health conditionalities themselves. For example, if increased income has a causal effect on such factors as reduced maternal depression, improved micro-nutritional quality of food intake, reduced family stress, and the purchase of educational materials and experiences, it is reasonable to expect that children's developmental trajectories will improve.

Figure 1 presents a simplified depiction of overall logic model for Family Rewards, showing the main processes through which the incentives payments are expected to reduce current and future poverty. A number of key assumptions and hypotheses underlying this model are important to highlight:

Improving children's educational outcomes

A growing body of evidence shows that poverty is associated with cognitive and academic problems among children. Poor children experience slower growth in vocabulary and other key indications of early language development. Indeed, extreme poverty (and exposure to violence) appears to constrain the optimal development of the underlying brain structure and processes central to early cognitive and language development and emotion regulation. Consequently, as early as kindergarten, poor children exhibit lower scores on tests of early literacy and math abilities as well as other indices of school readiness.³⁰ Left unaddressed, and compounded by attendance in lower-quality schools, these early disparities grow into the persistent achievement gap that much of educational policy and practice is now attempting to close.³¹ These problems contribute to the fact that poor children are twice as likely as their peers to repeat a grade and/or drop out of school before high school completion. According to longitudinal research that follows children through adolescence to adulthood, poor academic achievement and dropping out school are associated with worse future employment outcomes and earnings, thus contributing to the persistence of intergenerational poverty among many poor families.

For these and related reasons, children's school performance and academic achievement are a major focus of Family Rewards. Succeeding in school is one of the most important ways that children growing up in poor families can avoid being poor themselves in the future. Of course, school quality is obviously critical to student achievement and not directly addressed by Family Rewards. However, it is worth noting that the program is being launched at a time in New York City when efforts to increase school spending and school quality are also underway.

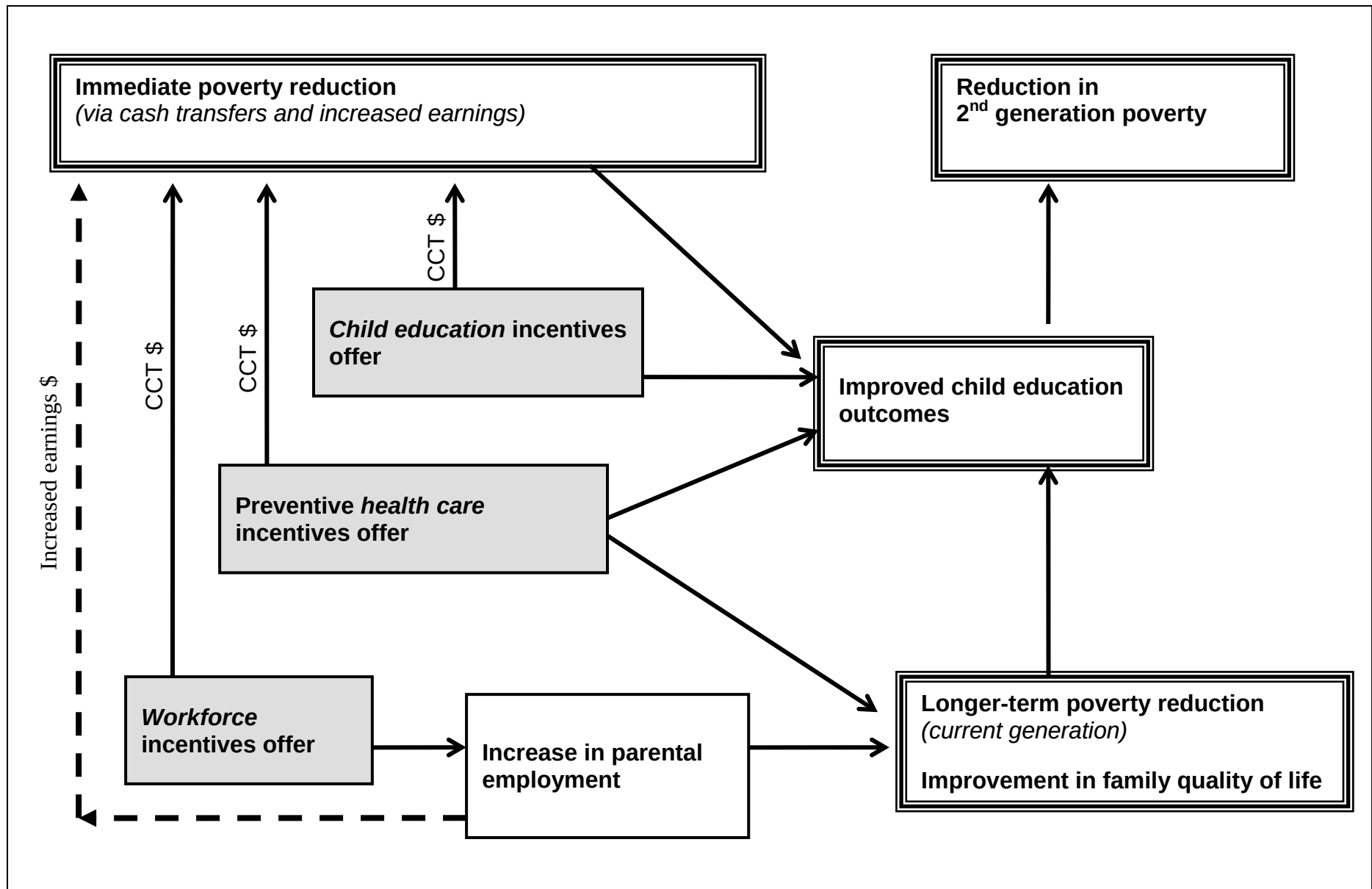
³⁰ Gershoff, 2003; Lee and Burkham, 2002; West et al., 2005.

³¹ Fryer and Levitt, 2004, 2005; Rathbun and West, 2004.

Opportunity NYC—Family Rewards

Figure 1

Theoretical Framework



To contribute to that success, the Family Rewards incentives offered in this domain are intended to encourage parents to become more directly engaged in their children's school performance, particularly at the elementary and junior high school levels. For example, it is hoped that the education-focused cash transfer will support and encourage parents to help their children strive harder in school, ensure that they attend school regularly, monitor and help them with their homework, and, when appropriate, seek other assistance that might improve their children's educational performance (e.g., tutoring or homework help or after-school programs). At the high school level, education incentives will be paid directly to the students, giving them an immediate financial stake in their own school engagement and performance.

Family Rewards may also boost some children's education attainment in other ways. For example, its health care component may promote early diagnosis and treatment of certain health and development problems that might otherwise make it difficult for students to sustain high attendance and to perform well in school, or for very young children to enter school ready to learn.

An increase in the family's short-term income—which may result from increases in increased parental earnings (see the dashed line in Figure 1) as well as from the direct cash transfers from all CCT payments—might also enhance children's school performance, as the evidence previously reviewed suggests, especially among younger children. (See the solid line in Figure 1 from immediate poverty reduction to improved child education outcomes.) Indeed, it may be that the Family Rewards education, health care, and workforce incentives will be more effective *together* in improving children's school outcomes than would the education-focused incentives if offered by themselves.

Improving family health outcomes

For a variety of reasons having to do with a lack of resources, knowledge, information, or other problems, many poor families often neglect routine preventive health care. Many do not maintain regular health insurance, either because they do not realize they are eligible for public insurance, do not follow through with paperwork and recertification requirements to continue receiving public insurance, cannot afford or choose not to pay the co-payment for employer-provided care, or are not eligible for either public or employer-provided insurance. Many also have no regular family physician and tend to rely on more costly hospital emergency rooms when they need care once a health problem arises. One study summarizing relevant literature on this topic states that, “There is substantial evidence that people who are insured are more likely than uninsured people to have a usual source of care other than the emergency room. Having health insurance and a usual source of care are generally among the strongest predictors of health services utilization, and they have been shown consistently to enhance timely use of medically necessary health services, increase use of preventive health care, increase continuity of care for chronic conditions, and reduce costly emergency room utilization.”³²

Not surprisingly, poor children tend to have worse health outcomes than their non-poor peers. These include higher rates of low birth weight, infant mortality, diarrhea, asthma,

³² Polit, London, and Martinez, 2001.

developmental disabilities, and environmental insults (e.g. lead poisoning).³³ Poor children are also twice as likely as non-poor children to experience an unmet medical need and more likely to receive pediatric care from hospital emergency rooms rather than pediatric practices. And despite the recent efforts to increase publicly-provided health insurance to children, 23 percent of poor children and 20 percent of near-poor children remain uninsured.³⁴

For all these reasons, increasing poor families' abilities and incentives to obtain and sustain adequate preventive health care for their children (and other family members) is a major focus of Family Rewards. The program hopes to achieve this in part by encouraging families to maintain health insurance (public or private), which, in turn, should encourage more preventive care and quicker responses to emerging health problems. In addition, the incentives are intended to promote regular, non-emergency health check-ups for all family members, and regular dental care. As previously mentioned, the benefits to improvements in preventive health care practices may include better school performance among children and better labor force participation among adults. In this regard, it is noteworthy that one ailment—asthma—is a leading cause of missed school among children and an important cause of missed work among adults. Moreover, studies of welfare recipients and other low-income populations suggest that health problems are a common impediment to steady work.³⁵

Finally, it should be recognized that, for all family members, good preventive health care practices may promote healthier lives by reducing unmet health needs. This would be a positive outcome for poor families even independent of any effects that better health outcomes might have for poverty reduction.

Improving parents' earnings

Family Rewards will increase family income through the direct cash transfers themselves—but only temporarily. *Sustained* reductions in child and family poverty after exiting the program will require that parents maintain regular employment. And, as previously mentioned, if increasing parental earnings boosts overall family income, the program's effects on young children's education and health may also improve.

For these reasons, the program includes a workforce component designed to promote steady, full-time work and the acquisition of skills to help participants qualify for better-paying jobs. It is believed that a CCT package that includes a workforce component for parents (with explicit incentives for human capital development) will be a stronger package overall for meeting shorter-term and longer-term, two-generation anti-poverty goals than would a CCT package that only focused on children's education and family health. Another consideration has to do with public acceptance: A cash transfer program that includes a component that explicitly supports and encourages families' pursuit of self-sufficiency may have broader appeal across the political spectrum than one that does not.

³³ Aber, Bennett, Conley, and Li, 1997.

³⁴ Gershoff, Aber, and Raver, 2003.

³⁵ Bloom, 1997; Polit, London, and Martinez, 2001; Zedlewski and Loprest, 2001.

It is true, of course, that for low-income families, work does not guarantee an escape from poverty. As Mayor Bloomberg's Poverty Commission report states, in 2005 over 46 percent of poor households in New York City were "working poor"—that is, they had incomes below the poverty level despite the fact that the head of household worked at least part of the year.³⁶ Skills deficits and wage stagnation are important factors contributing to this outcome. But while work is not always the remedy for poverty, it is also true that, for most families, it is impossible to improve family income and escape poverty *without* work. Indeed, it is noteworthy that among all New York City families, those without workers had a poverty rate of about 50 percent. In contrast, among all families with at least one worker, 11 percent were poor. Furthermore, only about 5 percent of all families in which the householder worked full-time, year-round were poor.³⁷ Poverty rates are much higher among female-householder families (with no husband present), although the patterns are the same.

Although steady work is important for improving family income, low-wage workers experience disproportionately high rates of job instability. According to one study, 60 percent of such workers aged 34 to 37 hold a job for at least a year.³⁸ This same study also points to research suggesting that job turnover among less skilled workers can negatively affect skills, wage levels, wage growth, and fringe benefits. Many welfare-to-work studies have also documented high rates of job turnover among single parents.

Family Rewards attempts to address this problem by providing incentives to sustain full-time employment, which, in some cases, will involve trading a part-time for a full-time job, or getting a new full-time job quickly, after a prior job ends. The program also rewards investments in training while employed, in the hope that this will help participants move into better paying jobs.

In the short-term, meeting the program's workforce-related conditionalities has the potential to boost family income substantially in three ways. Not only would family income increase directly from the Family Rewards workforce-based incentive payments, it would also grow from any extra earnings obtained through increased hours or duration of employment. In addition, these increased earnings could generate further income supplementation if the families take full advantage of the Earned Income Tax Credit (EITC), which they are entitled to receive. In addition, parents will be encouraged by the training incentives to build their own human capital through, for example, course-based skills training. This may position them to attain better-paying and more stable jobs in the future, and will be especially important once the families are no longer able to benefit from the conditional cash transfers. In fact, one might conjecture that any early effects of Family Rewards on children (due to the full package of incentives) might fade once the CCT payments end if families' incomes, material hardship, and financial stress return to their pre-program levels. Indeed, for many families, reducing the number of years that their children grow up in poverty or the severity of that poverty will depend far longer on their earnings capacity than their receipt of the CCT payments.

³⁶ CEO, 2006

³⁷ Based on data from the 2005 American Community Survey for New York City. ACS, 2005 Factfinder Table for NYC.

³⁸ Lane, 1999.

How incentives can influence behavior across the three domains

As some of the foregoing discussion suggests, conditioning income supplements on certain behaviors and achievements can influence family members in several ways. Some participants will view the cash transfer as an inducement to *adopt* new behaviors. For example, it might encourage some parents of elementary school children to begin checking that their children complete their homework assignments (to improve their school performance), finding a regular doctor, or looking for a full-time job. In other cases, the incentives may encourage parents to *sustain* or *intensify* positive efforts they already make—for instance, checking homework every night, getting all children to the doctor for regular annual check-ups, and finding a new full-time job more quickly once a prior job ends. In some cases, the incentive payments might function less as incentives (in the sense of motivators of behavioral change) and more as “enabling resources” to help families accomplish certain goals. So, for example, for parents in low-wage jobs who will lose pay if they take time off from work to bring a child to the doctor for a non-emergency check-up or to meet with a child’s teacher, the CCT payments offers some compensation for taking those actions. The same might be true for a parent who needs to pay for a babysitter for one child while taking another child to a free dental clinic across town, or to make it possible to attend a training course or to look for full-time work. Although the payments for any given activity are made after the fact, cash transfers obtained from compliance in one area (e.g., for health check-ups) may help facilitate achievement in another (e.g., paying a neighbor to pick a child up after school while the parent attends a course).

The cash transfers might also influence participants’ behavior in other indirect ways. For example, the incentives might encourage participants to get *services or assistance* (or *better* services) from existing programs and institutions in the community in order to increase their chances of meeting the conditions that are directly rewarded. As an illustration, the children’s education incentives may encourage parents to seek homework help or tutoring that might be available through the schools or non-profit organizations in their communities. Or it might encourage them to take advantage of One-Stop job centers (Workforce1) or job services offered by community-based organizations to get help finding full-time work or training opportunities.

The Family Rewards Incentives: Conditions and Amounts

As part of Family Rewards, families will be able to earn cash transfers that will largely be in the range of \$4,000 to \$6,000 per family per year, for two to three years. (This paper assumes that the program will offer incentives for three years, but this depends on funding availability. A final decision on the duration of the program has not yet been made.) The actual payment amount will depend on the degree to which families comply with the conditions that are set and, importantly, on the number of children in the family, each of whom can be a source of substantial transfer income. It is important to note that the proposed schedule of incentives is preliminary and that some important details of the plan remain to be specified.

Design considerations

In developing this schedule, the design team has been guided by the following key principles:

1. Within each domain, the conditions for incentives payments should be achievable with a reasonable level of effort.
2. Incentives should not be tied to services that are not generally available or reasonably accessible to the program group.
3. Within each domain, more money should generally be attached to conditions expected to be more challenging to meet.
4. The behaviors that draw payments must be verifiable in ways that are practical, timely, and resistant to fraud.
5. Incentives for children's school performance should avoid putting undue pressure on students or put them at risk of abuse if their family loses out on extra money because of their poor performance.
6. The amounts for any given activity should be substantial enough to appeal to families and encourage them to adopt and/or sustain the specified behaviors.
7. The amounts for any given activity, and overall, must not be so high as to be viewed as "unreasonable" by policymakers and the public, and, hence, politically unsustainable.
8. Full adoption of conditions for all family members across all domains should yield a total cash transfer that would amount to a substantial contribution to a participating family's budget (approximately 25%-30% of family income).
9. Recognizing that the greater the cost of the program per family, the fewer families that a replicated version of the program might be able to enroll, the total cash transfer per family should be at a level that is considered politically acceptable.
10. Because they are time-limited, the incentives should promote behaviors and achievements (i.e., build forms of human capital) that can become habits or that can be sustained or built upon after the cash transfers end.

It should also be noted that, for the demonstration, the cash transfers do not affect eligibility or payment amounts for most other types of transfers, including TANF, food stamps, Medicaid, SCHIP, housing assistance, or the EITC. (However, at this time, the CCT payments can affect a family's Supplementation Security Income payments.)

With these considerations in mind, and given the limited evidence base for knowing how different levels of payments might influence participants' choices and behaviors, the design team had to depend more heavily on insights and judgments than on behavioral science in setting payment amounts.

Table 5 summarizes the incentives schedule for the program. It lists the specific behaviors and achievements that earn payments, and the amount of those payments. In general, the payment amounts for the education and health components tend involve larger sums than has been true of most prior tests of incentives strategies in those two fields. This reflects, to some extent, the "immediate poverty reduction" goal of Family Rewards (i.e., the goal of relieving immediate hardship through a substantial cash transfer), not simply a judgment as to the minimum size of an incentive payment needed to induce behavior change. In contrast, the workforce incentives tend to be somewhat smaller than those tried in the largest demonstration projects that have tested or are currently testing wage supplementation strategies, but are still substantial. (When combined, the full set of transfers represents a very significant increase in the income of the very poor families)

Conditions and payment amounts

Part A of Table 5 presents the schedule of payments for children's education-related efforts and achievements. As can be seen, the education component focuses on an array of conditions that reflect educational achievement as well as efforts that can support good school performance. For all students, rewards are attached to excellent school attendance, obtaining a library card, parents' participation in parent-teacher conferences. For students in elementary and middle school (starting with children in grade 3), reward payments are offered for "good" performance on annual standardized tests in English Language Arts (ELA) test and math.³⁹ Payments are made if a student achieves a score that falls within the "proficient" range (a level 3 or 4 on a 4-point proficiency scale) or improves his or her score in a given year over the prior year by at least one full point on that four-point scale.⁴⁰ Additional rewards are offered for parents discussing the results of those annual tests with their child's teacher or school principal, and also for parents obtaining and reviewing on their own their children's scores on interim diagnostic tests that are administered five times over the course of the school year. (The last of these conditions will be dropped in the second year of the program due to changes being introduced by the Department of Education in its interim testing strategy, and also because of the general difficulty parents have in getting access to these scores, which are primarily intended for the teachers' use.)

³⁹ Recognizing that standardized tests are only given once a year, making the link between student efforts during the school year and the reward for those efforts remote, the design team considered trying to attach incentives to satisfactory completion of homework (as indicated on elementary school report cards), and improvement in grade point average. However, some reviewers strongly opposed these options, primarily because of inconsistencies in teachers' homework and grading standards, and concerns about conflicts that the incentives could generate between parents and teachers and parents and students if money were attached to those indicators of performance.

⁴⁰ Because standardized testing begins in grade 3, the opportunity to earn a reward through improvement from the prior year does not apply. It also does not apply to students who are new to New York City schools and have no prior-year standardized test score.

For high school students, Regents exams are the relevant tests, and students can earn a payment for passing each of the five core exams (i.e., scoring a grade of at least 65), an important graduation requirement in New York.⁴¹ High school students can also earn a small additional payment for taking up to two PSAT tests, which they can do without charge in New York City. Additional payments are offered to high school students who earn a minimally acceptable number of credits in a given academic year (11 each year toward to 44 credits needed to graduate.) This is to encourage students to pass all of their courses each academic year (taking advantage of summer school opportunities if necessary) so that they will remain on track to graduate in four years.⁴² An additional payment is offered for graduating.

As can be seen in the table, the payment amounts for attendance are higher for high school students than they are for elementary and middle school students. This reflects the fact attendance problems tend to be worse among high school students. Also important is the fact that, for elementary and middle high school students, payments will be made to the parents, since they usually expect to control the consumption behavior of their children at these ages. In contrast, for high school students, some payments will be made to the students themselves. This is because once students are in high school, they are more likely to care about the amounts of money attached to the incentives payments. They are at an age when they assume increased control over their personal consumption and, hence, are more likely to expect that their efforts should be rewarded by payments made directly to them rather than to their parents.

Only a small amount of money is attached to parents' attendance at parent-teacher conferences (\$25 twice per year). This is because these are generally brief interactions, and evidence of compliance of this behavior will come from a form signed by the teacher and submitted to the program by the parent, which is difficult to verify. Evidence regarding children's attendance and test scores will be obtained through electronic tracking data from the Department of Education.⁴³ (A later section of this paper discusses program procedures more fully.)

Part B of Table 5 presents the conditions and payment amounts proposed for the preventive health care component. This component includes activities pertaining to health insurance, preventive health care check-ups, and dental care. Many families who enroll in the study will be eligible for publicly provided health insurance through Medicaid, SCHIP, or Family Health Care Plus. However, it is well known that annual recertification requirements for these programs are onerous, and they make staying enrolled a complicated process for many

⁴¹ Special education students with individualized education plans (IEPs) would need to score 65 or higher on the Regents Competency Test, a simpler version of the Regents Examination.

⁴² A study of high school students in Chicago public schools found that attendance, course, grades, and credit accumulation in ninth-grade are correlated with the likelihood of graduating from high school. (Allensworth and Easton, 2007). Moreover, a New York City study of high school students highlights the problem of students quickly falling behind in their accumulated credits: "Overage and under-credited students fall behind early, and once they become off-track, they leave the system rapidly. 84% of students who are 16 years old with fewer than eight credits end up leaving the system." (Cahill, Hamilton, and Lynch, 2006.)

⁴³ For all grades, families of students with severe disabilities who are not able to take standardized tests can earn their rewards if the child completes an Alternative Assessment designed for students with disabilities. These assessments are tailored to individual education plans and assess progress in meeting specified functional outcomes using information collected over several months in a variety of ways.

poor families.⁴⁴ Consequently, “churning” on the rolls can be significant, with many otherwise eligible families losing their coverage. For example, one estimate within New York City suggests that 31 percent of Medicaid cases that did not complete a recertification and so were closed, but 27 percent of those closed cases were re-opened within 9 months.⁴⁵ The incentive payments are intended to encourage families to keep their coverage in place. A somewhat higher payment is offered to families where the parent is not eligible for public health insurance but has access to private, employment-sponsored health insurance. Some families may forego that insurance because of the co-payment costs. The CCT payments are intended to encourage those families to pay the co-payment and keep their health insurance in force. Other families that have incomes too high to qualify for subsidized care (for example, if their earnings grow after enrolling in Family Rewards), and may also find themselves in jobs that do not offer health insurance at all. For them, a higher incentive payment is offered to encourage them to enroll in other private group plans, but it is acknowledged that the cost of those plans may be prohibitive for most of these families.

The second set of health care incentives is designed to encourage families to get comprehensive, non-emergency physical examinations—the cornerstone of good preventive health care practices. Several purposes are envisioned here. One is to ensure that family members get medical assessments that can lead to early diagnosis and treatment of health problems that can become more serious over time. A second is to help educate them on health lifestyle (e.g., healthy eating, exercise, protection against sexually transmitted diseases). And a third is to establish a “medical home”—in other words, a relationship with a doctor or health care institution that knows their medical history and to which they can turn when problems arise, rather than resorting to hospital emergency rooms as a first response. To promote these visits, participants must present doctors with a special “preventive care checklist form” that identifies a set of common health conditions that doctors would be expected to explore or screen for in any thorough annual physical examination. This form is tailored to the different needs of adults, teenagers, and younger children. For infants and toddlers, the form includes a standard set of questions to encourage the doctor to screening for developmental problems and to make an appropriate referral for a fuller early intervention evaluation where warranted. An additional payment is offered to parents to follow-through with such a comprehensive evaluation (which is free in New York City) if they are advised to do so by their pediatricians.

The design team struggled with a practical way to create incentives for patients who need follow-up care to get it—without encouraging those who do not need it to try to get it, which would be an inefficient use of medical resources. The decision was to attach an incentive payment for one follow-up visit per family member per year (including for early intervention evaluations), in cases where the doctor has indicated the need for it on the initial health care checklist (and indicates the purpose and timeframe for making the follow-up visit on a subsequent form). The payment amount for the second visit is half that of the first visit, to help temper the incentive to seek unneeded medical care.

⁴⁴ TANF and Safety Net Assistance (SNA) recipients are automatically enrolled in Medicaid and need not reestablish their eligibility as long as they remain on TANF or SNA. Therefore, Opportunity NYC participants will not be eligible for the program’s health insurance incentive while they are TANF or SNA recipients.

⁴⁵ To be verified with HRA.

Finally, the health component includes incentives for preventive dental care—that is, for regular cleanings and check-ups. It is recognized that many dentists do not accept Medicaid. However, there are a number of dental clinics around the City that offer free or reduced-cost dental care. Finding them, getting to them, and paying for them (if they charge a fee) are likely to be a burden. It is hoped that the incentives payments (which could, of course, be used to help cover the out-of-pocket cost of those visits) for dental care will encourage families to take on those extra burdens.

Part C of Table 5 presents the workforce component, which is aimed at the parents, and which has two main features. The first is a payment for sustained full-time employment. Operationally, this means that a participating parent must work at least 30 hours per week for six out of every eight weeks. Allowing for some “downtime” is a way of recognizing that, for many low-wage workers, job turnover is common, sometimes because the job itself ends. Those who are in this situation or who leave work for other reasons would have a strong incentive to seek another full-time job quickly. This payment is modeled after the approach being tried in the UK ERA demonstration described earlier, and also builds on strategies tested in a number of other “make-work-pay” evaluations for welfare recipients.

The workforce incentives also incorporate payments for approved education and training activities that can help build participants’ human capital so that they can qualify for higher-skilled and better-paying jobs. The courses may be shorter-term or longer-term, and the incentive payments are tailored with that in mind—for example, providing a higher payment for a longer-term course. Instruction may include not only specific occupational skills training, but also ESL instruction and Adult Basic Education and GED preparation. Importantly, payment is contingent on completing the course or (in the case of ESL and basic education) reaching a particular stage or proficiency level.⁴⁶ In addition, the payment is only available if the participant is also working at least 10 hours per week. The rationale for this is to discourage participants from remaining or dropping out of the labor force in order to undertake training, which would be inconsistent with City’s welfare-to-work policies. Moreover, for participants not on welfare, who have few other resources on which to live, training in the absence of work is a luxury most would not be able to afford.

In addition to incentives tied to activities and accomplishments in the education, health, and workforce domains, Family Rewards offers a one-time \$50 incentive payment for opening up a new bank account (or using an existing account) into which reward payments can be deposited electronically.

⁴⁶ In order to earn the payments for these activities, participants in adult basic education, GED preparation, or ESL classes, for which standards of completion are often ambiguous and compliance hard to measure, must provide documented evidence from their providers indicating that they have made satisfactory progress in their classes, and that they have participated for the required number of hours established by Family Rewards for a given level of payment.

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Table 5

Tentative Schedule of Incentives Payments, by Type of Activity

A. Children’s Educational Efforts and Achievement
Grades 1-8 (<i>Payments made to parents</i>): • <u>Attendance</u>: \$25 per child per month (max: \$250 per year, covering 10 months of school) for satisfactory attendance (95% of scheduled days, with provision for extended illness). • <u>Parent-teacher conferences</u>: \$25 per conference, 2x per year (max: \$50 per year per child) for parent’s attendance at parent-teacher conferences. • <u>Library card</u>: \$50 paid once during program if child gets (or has) a public library card. • <u>Reviewing results of low-stakes interim tests</u>: \$25 for parents to acquire and review on their own their children’s performance on interim standardized tests intended to help teachers diagnose students’ progress (up to 5 times per year; max: \$125 per year per child). (To be dropped in second year of the program.) • <u>Test scores (starting in grade 3)</u>: – For grades 3-5: \$300 per child for scoring at a level 3 (indicating proficiency) or above on standardized ELA test, or (starting in grade 4) for improving by at least 1 level over prior year’s level; same for standardized math test. (Max: \$600 per year per child.) – For grades 6-8: \$350 per test for meeting the same conditions. (Max: \$700 per year per child.) • <u>Discussing results of annual ELA and math tests with school (starting in grade 3)</u>: \$25 per test, 1x per year (max: \$50 per year per child) for parents to discuss child’s test results with teachers or principal and get confirmatory signature.
Grades 9-12 (<i>Payments split between parents and students, as indicated below</i>): • <u>Attendance</u>: \$50 per child per month (max: \$500 per year, covering 10 months of school) for satisfactory attendance (95% of scheduled days, with provision for extended illness). (<i>50% paid to student, 50% paid to parent.</i>) • <u>Parent-teacher conferences</u>: \$25 per conference, 2x per year (max: \$50 per year per child), for parent’s attendance at parent-teacher conferences. (<i>100% paid to parent.</i>) • <u>Library card</u>: \$50 one-time payment if child gets (or has) a public library card. (<i>100% paid to student.</i>) • <u>Test scores</u>: \$600 per child for passing (i.e., scoring 65 or above on) each of 5 Regents tests (max: \$3,000 during program) (<i>100% paid to student.</i>) • <u>Credit accumulation</u>: \$600 per year per child for accumulating 11 credits during a school year. (<i>50% paid to student, 50% paid to parent.</i>) • <u>PSAT</u>: \$50 for taking PSAT test up to 2 times (max: \$100 during program). (<i>100% paid to student.</i>) • <u>Graduation</u>: \$400 payment for graduating high school. (<i>50% paid to student, 50% paid to parent.</i>)

Continued

Table 5 (continued)

B. Family Preventive Health care Practices
Maintaining health insurance: • \$20 per month (max: \$240 per year) for each parent for maintaining public health insurance (including Medicaid and Family Health Plus coverage) for each parent. • \$20/month (max: \$240 per year) for maintaining Medicaid or SCHIP coverage for all children (together). <i>[Not for TANF recipients due to automatic Medicaid enrollment.]</i> • \$50/month (max: \$600 per year) for each parent for maintaining private/employer health insurance for each parent. \$50 per month (max: \$600 per year) for maintaining private/employer insurance for all children (together). Non-emergency health screenings and early intervention: • <u>For adults and children:</u> \$200 per family member per year for completing an annual non-emergency medical check-up. Physician must fill out “preventive health care form” indicating that a minimum set of age-appropriate screenings and assessments were conducted and that other health information was reviewed with the patient and/or parent. \$100 per family member per year for completing a physician-advised follow-up visit within a specified timeframe. • <u>For young infants and toddlers (children under 30 months of age):</u> \$200 per child for completing a pediatrician-advised early intervention evaluation. • Dental care: \$100 per family member for cleaning and check-up, 2x per year for ages 6+ and 1x per year for ages 1-5.
C. Adult Workforce Efforts
Sustained full-time employment: • \$300 for for working full-time (at least 30 hrs/week for 6 or more weeks in each 2-month payment period—i.e., approximately 75% of the time) (max: \$1,800 per year @\$150/month) Education and training while employed: • Payments for completing an <u>approved</u> education or training course while holding a job. Must work at least 10 hrs/week while attending course. \$300 per each course lasting 35-70 hours; \$400 per each course lasting 71 to 140 hours; \$600 for each 141 hour-increment of a course lasting more at least 141 hours (max: \$3,000 per adult during program). (Training may include ESL, basic skills, and GED prep courses.)

Variation in total payment amounts by family type

The fact that Family Rewards is a full-family, two-generation model with incentives tied to efforts and outcomes for each family member means that the maximum cash transfer that families can receive under the program will vary considerably by family size. Table 6 illustrates this variation by constructing six hypothetical families with different numbers of children, and with children in different grade levels. The table assumes that each family will earn all or a very large proportion of the incentives payments available to it. Thus, it shows the substantial amounts of money that a family *could* receive in reward payments. What they actually receive may be lower, of course, depending on their levels of compliance. As the table indicates, a single mother with one child in fourth grade (Family #1) could earn a total cash transfer of \$4,405 per year for meeting all of the education and health care conditions and working full-time. A single mother with two children, in grades 5 and 9 (Family #3) who did the same could earn even more—up to \$6,995. A single mother with three young children of various ages who combined part-time work with training (Family #4) could earn \$6,310. Two-parent families with two working parents and several children could earn particularly large sums. For example, such a family with children in the fourth, seventh, and ninth grades, and in which one parent works full time and the other works part time and receives the entire annual training bonus (Family #5), could earn \$10,300. It should be noted, however, that two-parent families with both parents enrolled in Family Rewards make up only about 6 percent of all families in the program.

To put these numbers into some perspective, it is useful to compare them to other benchmarks. One is the federal poverty level, which varies with family size (see Table 7, column 5). Another is the total amount of money that would be earned by a person who worked 40 hours per week at the current New York State minimum wage (\$7.15 per hour), which would total \$14,872 per year (see Table 7, column 6). For Family #1, the total Family Rewards payment would amount to 31 percent of the poverty level for that family and 30 percent of the mother's annual minimum-wage earnings. For Family #3, the Family Rewards payment would amount to 40 percent of the poverty level and 47 percent of annual minimum-wage earnings.

Of course, many families will earn only a portion of the payments available to them. Nonetheless, even at partial payment, families stand to earn substantial transfers. For example, if Family #1 actually earned only half of the total Family Rewards payment shown in Table 7, it would receive \$2,202 (or 16 percent of the poverty income threshold for such a family—not shown in table). And if Family #3 earned half of the CCT estimate, it would receive \$3,497 (or 20 percent of the poverty income threshold).

Share of total payments by CCT component

Table 6 also illustrates, for the same group of hypothetical families, the proportion of the total CCT payment accounted for by each of the three program components. Again, this distribution will vary by family type. For example, in Family #1 (a full-time working single parent with one elementary school student), the education component accounts for only 20 percent of the total payments. However, it accounts for 37 percent for Family #3 (a similar family but with two older children) and 49 percent for Family #6 (a similar family but with two high school students). Similarly, across the families illustrated in Table 6, the health care

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Table 6

Illustration of Potential Annual Payment Amounts by Type of Family

Family #1: Mother and 1 child in grade 4		
Incentives	In Dollars	Percentage
Maximum Education Payments		
Child 1:	\$1,125	
Subtotal:	\$1,125	26%
Maximum Health Payments		
Adult 1:	\$740	
Child 1:	\$740	
Subtotal:	\$1,480	34%
Maximum FT work bonus	\$1,800	41%
Training bonus	\$0	0%
TOTAL PAYMENTS	\$4,405	100%

Family #2: Mother and 2 children in grades 4 and 7		
Incentives	In Dollars	Percentage
Maximum Education Payments		
Child 1:	\$1,125	
Child 2:	\$1,225	
Subtotal:	\$2,350	37%
Maximum Health Payments		
Adult 1:	\$740	
Child 1:	\$740	
Child 2:	\$740	
Subtotal:	\$2,220	35%
Maximum FT work bonus	\$1,800	28%
Training bonus	\$0	0%
TOTAL PAYMENTS	\$6,370	100%

Family #3: Mother and 2 children in grades 5 and 9		
Incentives:	In Dollars	Percentage
Maximum Education Payments		
Child 1:	\$1,125	
Child 2: (incl. 1 Regents)	\$1,850	
Subtotal:	\$2,975	43%
Maximum Health Payments		
Adult 1:	\$740	
Child 1:	\$740	
Child 2:	\$740	
Subtotal:	\$2,220	32%
Maximum FT work bonus	\$1,800	26%
Training bonus	\$0	0%
TOTAL PAYMENTS	\$6,995	100%

Family #4: Mother and 3 children, 1 age 3, 1 in grade 2, 1 in grade 4		
Incentives	In Dollars	Percentage
Maximum Education Payments		
Child 1:	\$0	
Child 2:	\$1,125	
Child 3:	\$1,125	
Subtotal:	\$2,250	36%
Maximum Health Payments		
Adult 1:	\$740	
Child 1:	\$640	
Child 2:	\$740	
Child 3:	\$740	
Subtotal:	\$2,860	45%
Maximum FT work bonus	\$0	0%
Training bonus	\$1,200	19%
TOTAL PAYMENTS	\$6,310	100%

Family #5: Mother, Father, 3 children in grades 4, 7, and 9		
Incentives:	In Dollars	Percentage
Maximum Education Payments		
Child 1:	\$1,125	
Child 2:	\$1,225	
Child 3: (incl. 1 Regents)	\$1,850	
Subtotal:	\$4,200	41%
Maximum Health Payments		
Adult 1:	\$740	
Adult 2:	\$740	
Child 1:	\$740	
Child 2:	\$740	
Child 3:	\$740	
Subtotal:	\$3,700	36%
Maximum FT work bonus		
(assumes 1 parent works FT, 1 parent works PT)	\$1,800	17%
Training bonus	\$600	6%
TOTAL PAYMENTS	\$10,300	100%

Family #6: Mother and 2 children in grades 9 and 11		
Incentives:	In Dollars	Percentage
Maximum Education Payments		
Child 1: (incl. 1 Regents)	\$1,850	
Child 2: (incl. 3 Regents)	\$3,050	
Subtotal:	\$4,900	55%
Maximum Health Payments		
Adult 1:	\$740	
Child 1:	\$740	
Child 2:	\$740	
Subtotal:	\$2,220	25%
Maximum FT work bonus	\$1,800	20%
Training bonus	\$0	0%
TOTAL PAYMENTS	\$8,920	100%

Notes: Maximum health payments assume that families receive public health insurance.

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Table 7

CCT Payments as Percentage of the Federal Poverty Line and Annual Minimum-Wage Earnings

Family	Annual FPL	Annual Min. Wage Earnings	Projected CCT Payment	CCT as % of FPL	CCT as % of Annual Min. Wage Earnings
Family #1	\$14,000	\$14,872	\$4,405	31%	30%
Family #2	\$17,600	\$14,872	\$6,370	36%	43%
Family #3	\$17,600	\$14,872	\$6,995	40%	47%
Family #4	\$21,200	\$14,872	\$6,310	30%	42%
Family #5	\$24,800	\$22,308	\$10,300	42%	46%
Family #6	\$17,600	\$14,872	\$8,920	51%	60%

component accounts for 24 percent to 47 percent of the total CCT payment, and the workforce component (combining the full-time work and training-related bonuses) accounts for 22 percent to 50 percent.

Delivering the Incentives—and Information

Seedco has primary responsibility for operationalizing the delivery of the CCT payments. This entails reviewing all documentation, verifying compliance with CCT criteria, authorizing payments, and contracting with a financial institution partner to transfer payments electronically to participants' bank accounts. As mentioned previously, Seedco also contracts with six Neighborhood Partner Organizations (NPOs) that helped to enroll eligible families into the study and now are an important ongoing source of information and support to participants who seek guidance on services that might help them complete the activities that can earn them reward payments. In addition to this local assistance, information on service opportunities and customer support in regard payment inquiries and problems, is provided centrally through helpline call-in number staffed by Seedco personnel. Participants can obtain additional information from a program Web site.

Overview of key steps in the process

The following steps and processes illustrate how the Family Rewards incentive system works (see later section on Participant Recruitment, Random Assignment, and Flow for further details):

- After enrollment, families selected to be in the program group are told about the program and what they need to do to earn the incentives.
- Payments are made on a bi-monthly basis.
- Payments are made for any given milestone achieved during a given payment period, after verification that a family has met the required conditions. Seedco calculates the amounts owed to each family and then authorizes a subcontracted financial institution to make the payment.
- Payments are typically made electronically to participants' bank accounts (new or existing). Participants who chose not to open a new account or accept program payments into an existing account are paid through a stored value card. Those who accept the bank account option get an additional one-time \$50 bonus payment for having an account. New bank accounts established for the program⁴⁷ come with "overdraft-impossible" debit cards to help participants avoid incurring overdraft charges and slipping into debt. These accounts also have no minimum-balance requirements or fees. The extra incentive tied to maintaining a bank account recognizes the consumer value to participants of establishing a relationship with mainstream banking institutions for basic financial services as an alternative to expensive neighborhood check-cashing outlets. It also makes the program less costly and more efficient to administer than if payments were made by checks.
- Seedco mails earnings statements to program participants after each bi-monthly payment is made, informing participants of how much money they earned during that period.

Tracking and verifying compliance

Accurately and swiftly determining when a participant has met the conditions that earn an incentives payment is crucial to the smooth operation of the overall incentives system. It is important not only for maintaining the program's credibility in the eyes of the participants, but also for avoiding fraud and maintaining the program's integrity in the eyes of policymakers and the broader public. It is also important to provide participants with information that promotes full compliance with conditions. The verification process differs depending on the types of activities involved. For example, for school performance outcomes, Seedco obtains administrative records data in electronic form from the Department of Education on a regular basis. Similar, electronic data matches with the Human Resources Administration/Department of Social Services are used to verify maintenance of public health insurance. For other activities, participants are supplied with individually customized "coupon books" that include preprinted forms that participants mail-in with supporting documentation (e.g., pay stubs for full-time work, forms signed by

⁴⁷ With the assistance of the city's Office of Financial Empowerment, several banks and credit unions have created special Opportunity NYC savings accounts for program participants.

doctors verifying medical visits, forms signed by teachers verifying attendance at parent-teacher conferences or discussions of annual test scores, and so on). The coupon books also include preprinted forms reminding them of the activities that are auto-verified. Quality control and audit procedures, based on established quality assurance protocols, have been instituted to ensure the integrity of this reporting process.

Information and communication component

It is anticipated that many parents who enroll their families in Family Rewards will embrace the program's goals but need assistance in order to succeed. For example, many may not know how to find a regular family doctor, how to help their children perform better in school, or where to get help finding a job or training. They may also have questions about the mechanics of the program (e.g., "how do I submit documentation?" or "how often will I get paid?") The Information and Communication component of the program is intended to address this need. This component will be operationalized in four major ways:

- ***Ongoing marketing of opportunities to participants.*** Seedco coordinates an intensive communications strategy, largely through the mail, to remind participants about the incentives offers and what they can do to earn the money the program is making available.
- ***A customer service call-in center or "helpline."*** This service, operated by Seedco and its partners, allows participants who call a dedicated telephone number to speak to a customer service representative who can provide participants with help regarding past or upcoming reward payments, as well as how to document their compliance with the program's conditionalities.
- ***Walk-in services and workshops at the Neighborhood Partner Organizations.*** The NPOs are also available to families to get assistance that can help them maximize their reward payments. The NPOs provide the following assistance: (1) **workshops** held periodically to review program processes, guidelines, and address special topics in each of the program's major domains (education, health, and workforce); (2) **service referrals** from staff who can direct individuals/families to service providers and programs in the community that may help them earn rewards (e.g. tutoring programs, health clinics, job centers); and (3) **in-person customer support** to meet with program participants to assist them in gathering required documentation as well as to troubleshoot problems related to documentation or payment.
- ***Program Web site.*** Seedco established a Web site with information about the program, guidelines for documentation, and information on service agencies available to residents of each community.

The Target Group Children and Sampling Strategy

Only families with children entering one of the three target grades (fourth, seventh, and ninth) in the fall of 2007 were deemed eligible for the study. This was done to ensure that a large enough sample of children would be available to support a separate analysis of the

program’s impacts on educational outcomes for each of those groups. However, once a family enrolled in study, all children in the family became part of the evaluation. Among the children of program-group families, all siblings of the target-group children who are 18 years old or younger are eligible for the program’s rewards and will be included in the analysis.

As previously mentioned, the study sample includes 4,778 families (with 5,051 adults and 11,489 children), randomly allocated (evenly) to the program group or control group. Most of these families were recruited during July through December, 2007. As the box below shows, over 1,800 students are included in the fourth and seventh grade cohorts, while over 2,000 are included in the ninth grade cohort. Although these cohorts will be the main focus of the study’s assessment of the program’s impacts on children’s educational outcomes, parts of the analysis will also focus on children in other grades.

Number of Children in Each Target-Grade Cohort <i>(Program and Control Groups Combined)</i>		
Cohort	Grades in Which Incentives Are Available	Number of Children <i>(Including siblings)</i>
Cohort A	Start as 4th graders	1,881
	Continue as 5 th graders	
	Continue as 6 th graders	
Cohort B	Start as 7th graders	1,817
	Continue as 8 th graders	
	Continue as 9 th graders	
Cohort C	Start as 9th graders	2,098
	Continue as 10 th graders	
	Continue as 11 th graders	

- **Why fourth, seventh, and ninth grades?**

The fourth, seventh, and ninth grades are widely believed to be at or near the start of critical educational transition years for students. It was thus considered a high priority to determine whether the Family Rewards education incentives could make a difference for students entering the study at those grade levels, and to ensure that sample sizes would be large enough at each of those levels to permit a statistically rigorous and independent impact assess for each of those groups of target children.

The three grade levels confront students with different types of educational challenges. For example, by fourth grade, children are making a transition from “learning to read” to “reading to learn”—in other words, applying their newly developed reading skills to acquire content knowledge. Without making that leap well, future school work becomes more difficult. In general, children who fall seriously behind in educational performance by the third or fourth grade tend to have difficulty catching up later.⁴⁸ It is hoped that initiating a three-year incentives program for the fourth-grade target group will help many children in that group do better in that critical year. In addition, for fourth-graders, reward payments based on test scores can be tied to

⁴⁸ For one study showing this pattern, see Clotfelter, Ladd, Vigdo, 2006.

improvements from scores achieved in grade 3 (the first year that standardized tests are administered system-wide in New York City), not just to meeting proficiency standards in grade 4. Beginning incentives in the fourth grade also has the advantage of continuing them (if they last for three years) through the first full year of middle school, which typically begins in grade 6 in New York City, and is a crucial transition year. Similarly, initiating the incentives program for a cohort of seventh-graders is intended to help junior high school students perform solidly during those important years and through ninth grade (the first of high school), another critical turning point for students.

While the seventh-grade cohort will see its incentives end after ninth grade, the ninth-grade cohort will just be starting its three-year incentives program in grade 9 and continuing through the end of grade 11 (if the program lasts for three years). Many ninth-grade students in poor communities perform poorly in that grade, begin to fall seriously behind in credits, and begin patterns of poor attendance, all of which puts them on a slippery slope toward dropping out of school. Students who succeed through the tenth and eleventh grades, however, are highly likely to graduate. Thus, offering students education-focused incentives beginning in the ninth grade—which many educators view as a “make-or-break” year for high school students—may help boost achievement at this very critical stage; continuing incentives for one or two more years may help sustain their engagement and performance in school, putting them on a solid path toward graduation.⁴⁹

- **Recruiting from school lists**

As described more fully below, families potentially eligible for Family Rewards were identified from lists of students provided by the New York City Department of Education. These lists identified not only students in the target grades and living in relevant community districts, but also those students enrolled in the free school lunch program. The free school lunch indicator provided a practical way to identify students whose families were likely to meet the income eligibility criterion for the Family Rewards, which is the same as that used for food stamps and certain other government benefit programs—i.e., family income that is at or below 130 percent of the federal poverty level.⁵⁰

The advantage of drawing most sample members from school lists of potentially eligible students and their families is that it will allow the analysis to speak to issues concerning the generalizability of the impact findings to a definable larger population. However, because

⁴⁹ Because one cohort in the study (Cohort B) will see its incentives offer ending in ninth grade, while another (Cohort C) will see it just beginning in ninth grade, the analysis can explore the relative value of each of those different starting points for subsequent high school performance.

⁵⁰ It is recognized that this is an imperfect way of identifying families with incomes at or below 130 percent of the poverty line. For example, some students from low-income families who might be eligible for the free school lunch program might never have applied for it and would thus have missed the chance to be recruited for Family Rewards, while other families may have no longer been receiving free school lunches by the time they enrolled in the Family Rewards study. Moreover, schools do not require the same level of detailed documentation of family income to determine eligibility for the free school lunch program that other benefit programs, such as food stamps and TANF, require, allowing for the possibility of more error in establishing a family’s actual income. (Families establish eligibility for free school lunches by completing relatively simple enrollment forms that are distributed by the schools.) However, given the generally high poverty rates of the families living in the targeted community districts plus the complexity of building a new income verification system just for this demonstration, it was decided that the free school lunch indicator would be a satisfactory basis for identifying low-income families appropriate to include in this study.

targeted families must volunteer for the study and sign informed consent form, those who do volunteer will not perfectly represent the larger school-list population. Still, they are expected to be substantially similar. In fact, a preliminary analysis of prior-year test standardized test scores shows that the likelihood of scoring at a level deemed proficient (i.e., a level 3 or 4 on a 4-point scale) was nearly the same for students in the research sample compared to all students on the larger recruitment list from which they were drawn.

Although the sample selection criteria were driven primarily by the desire to ensure that there would be enough students in each of the three grade-specific cohorts to permit a thorough grade-specific impact analysis, this sampling strategy is also suitable for other aspects of the impact analysis that do not directly concern children's educational outcomes. For example, for assessing the effects of Family Rewards on family health-related outcomes, adult earnings, family income, progress in moving out of poverty, and so on, the study will benefit from having a sample of families that covers a wide spectrum of children's ages. In other words, the sample includes families with young children (i.e., elementary school children), pre-teens and young teens (i.e., junior high school children), and older teens (i.e., those in high school). It will also include some parents with pre-school-age children who are siblings to school-age children. This range of families is important because family members' health care needs and parents own employment and training choices, and, consequently, how they respond to the health care and workforce components of Family Rewards may vary substantially with the ages of their children.

It should also be noted that although the families were only eligible for the study if they were living in designated community districts, their children were eligible for the study even if they attended schools located outside of their target community districts.

Participant Recruitment, Random Assignment, and Flow

The key steps for identifying eligible families, enrolling them into Family Rewards, and assisting them in the program are summarized in Figure 2. The major steps include the following:

- ***Securing lists of eligible families.*** As previously described, in order to be eligible for Family Rewards, families in the six community districts had to have a child in the fourth, seventh, or ninth grade in September 2007. They also had to have incomes at or below 130 percent of the federal poverty level. Parents of children in targeted grades were identified from school lists provided by the New York City Department of Education (DOE), which also indicated whether the students in those grades were eligible for free school lunch (the basis for determining whether families had incomes below the eligibility cutoff level.)
- ***Sampling families from the lists of eligibles.*** Families on the DOE school lists who met the student grade and income criteria, made up the primary sampling frame for recruiting study participants. From this population, MDRC randomly selected a series of subsamples or batches of families from each community district, with approximately 200 families per batch per district. Once a given batch was selected, contact information for each family was provided to Seedco. Seedco and its Neighborhood Partner Organizations then attempted to get as many of those families as possible to volunteer for the study. As previously described, parents (or guardians) enrolling in the demonstration had to undertake a special screening and enrollment

process, sign an informed consent agreement to allow access to certain personal data for evaluation purposes, and submit to random assignment procedures that determined whether they were assigned to the program or control group. Once Seedco and its partners made a substantial effort to recruit volunteers from a given batch, it was provided with a new batch. This process was repeated until the sample intake period ended (in January 2008).⁵¹

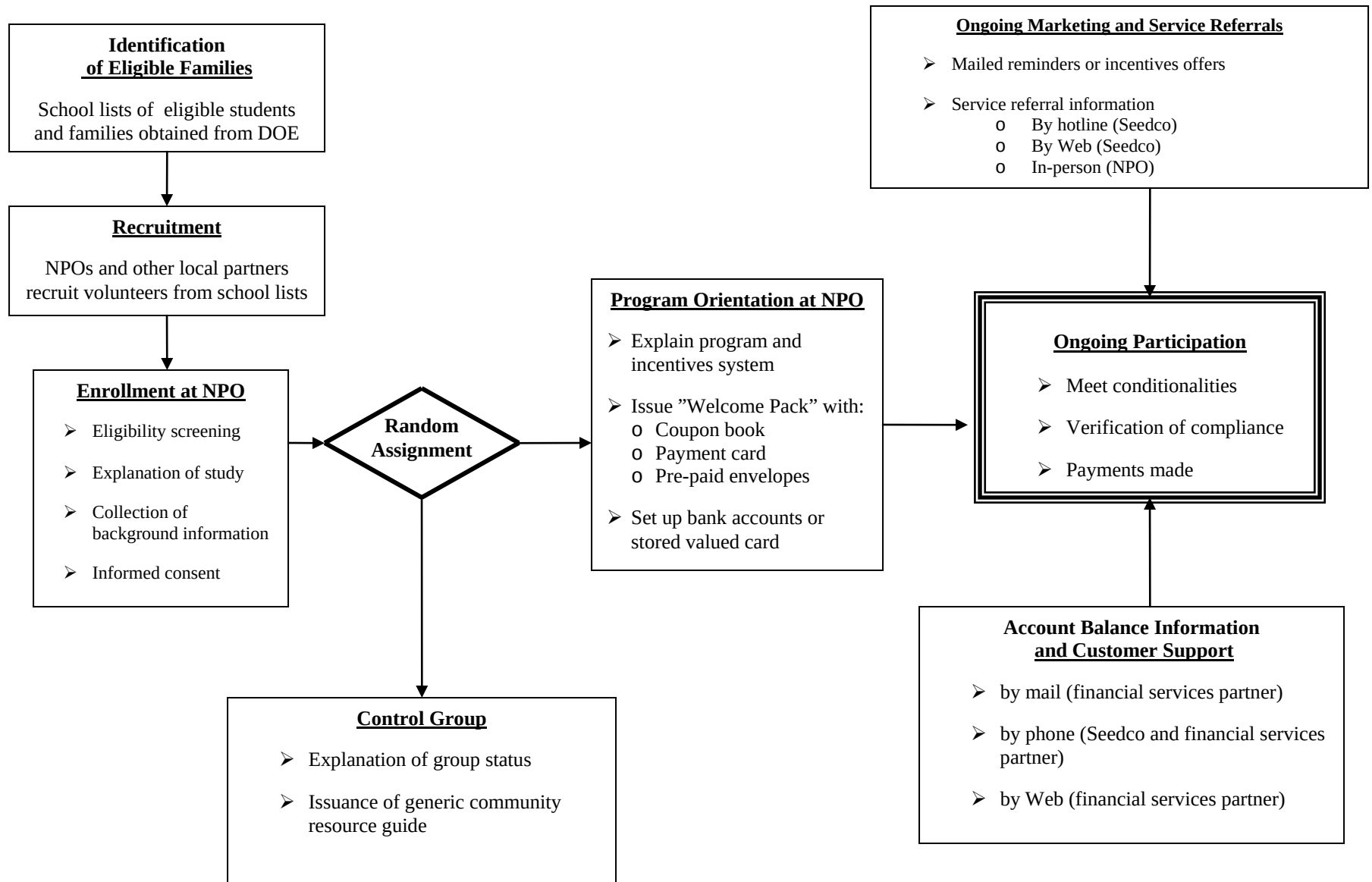
- ***The outreach and recruitment process.*** Seedco worked with NPOs to conduct outreach to families who participated in the Family Rewards program. A variety of methods were used, ranging from mailings and fliers to calling and contacting people in-person at their homes. In many cases the initial contact information provided by the school lists was incorrect or outdated, so NPO recruiters often had to seek updated contact information by visiting neighbors near the last known address and inquiring about the families' whereabouts.⁵² The NPOs also partnered with other local organizations and establishments, including schools, clinics, churches, etc., to try to reach eligible families. Contacted families interested in participating in the program were invited to come to the NPO to learn more about the Family Rewards program and possibly enroll in the study.
- ***Eligibility screening, enrollment and random assignment.*** Individuals interested in participating in the program met individually with intake staff from the NPOs who collected demographic and other background information and to determine eligibility, entering all information into an online data system developed by MDRC. Individuals consenting to enroll and participate in the program were randomly assigned to program or control groups as determined by the MDRC system.
- ***Informing enrollees of the random assignment outcomes.*** All families who enrolled in the program and subsequently took part in the random assignment process were informed of their status by mail through letters sent by Seedco (to the program group) or MDRC (to the control group). Families assigned to the control group were reminded that the decision was based purely on chance and had nothing to do with them personally. They were also given a generic list of service agencies in their community that might be helpful to their families, which they were free to contact.⁵³ They were also told that they might be contacted by the survey firm to be interviewed for the evaluation, and if they completed an interview, they would be paid a stipend for

⁵¹ While the random assignment process officially ended in January 2008, several additional enrollees entered the study in March 2008 due to inadvertent delays in processing their applications and determining their eligibility.

⁵² Further details about the recruitment experience will be presented in a future report on the implementation of the Family Rewards program.

⁵³ By virtue of their participating in the random assignment process, families who became controls were, of course, exposed to neighborhood-based service institutions and were also given a generic list of services that could help them improve the kinds of outcomes that Family Rewards is promoting. The exposure to this information could increase the control group's own service use and anti-poverty efforts (if they choose to act on the information) beyond what it would have been in the absence of the study. However, it is expected that any such effect will be small and will not undermine the main treatment intervention being tested, since the control group is not eligible for any special incentives payments to encourage greater service receipt. Nonetheless, if the control group's service use were increased, and if those services were effective, the estimated impacts of Family Rewards would be smaller than they would be otherwise.

Figure 2
Participant Flow



- their time. As an expression of appreciation for the time that all families took to enroll in the study and undergo random assignment (whether they ended up in the program or control group), the families were given a \$24 Metrocard for use with the City’s subway and bus systems.
- ***Program Orientation.*** Families assigned to the program group were scheduled for a program orientation session at which they were provided with the following information and materials: the list of incentives and how they could earn them; documentation submission procedures and the payment schedule; policies for verification and payment; information for the “helpline” and other CCT support services; and information on available programs and services to help meet the conditions that would earn reward payments. However, due to the quick start-up of the demonstration, final versions of documents and forms were not available at the time of the initial orientation sessions, each family was eventually provided with a customized welcome packet that included documentation coupons, pre-addressed and pre-paid envelopes for documentation submission. Families were also given instructions on opening Opportunity NYC bank accounts (and or using stored value cards), or how to arrange for direct deposits of program payments into a family’s existing bank account.
- ***Ongoing information and communication.*** An absolutely critical, ongoing operational mission will be to get very clear, strong and repeated information to participants explaining what they can do to maximize payment amounts. Thus, regular communication will be essential throughout the term of the program. As previously mentioned, financial statements are sent to participants with payment information and a summary of incentives conditions that have been met. Regular mailings will also describe service offerings available to participants in their communities (e.g., informing them where they can get help with education, health care, and workforce issues), and reminding them that they can visit or call the NPOs to get additional help finding appropriate services.

Evaluation Overview

The demonstration will include a comprehensive evaluation of Family Rewards, involving 4,778 volunteer families, with 2,388 randomly assigned to the program group and another 2,290 assigned to the control group. The evaluation will have three major strands of research: an impact analysis, an implementation and process analysis, and a benefit-cost analysis, which are described below.

Analysis strands

- **Impact analysis.** This analysis will examine the effectiveness of Family Rewards in improving outcomes for the families participating in this CCT initiative. Put simply, it will determine whether the program group had better outcomes than it would have achieved without the program. The study will track both the program and the control

groups for a number of years using administrative and survey data to measure outcomes. Because random assignment, when properly implemented, helps eliminate systematic differences between the program and control groups prior to the start of the program, any subsequent differences in outcomes—for example, differences in school performance among children, and differences in family income and poverty—can be attributed to the program with confidence. The impact analysis will examine the program’s effects on a wide range of outcomes, including children’s school performance; family health care practices and health outcomes; parents’ employment and training outcomes; and family income, benefit receipt, poverty, material hardship, and quality of life. The impact analysis will assess the program’s effects on families covering a period of at least five years after each family’s date of random assignment. If the program intervention lasts for three years, the evaluation will have an opportunity to assess whether the impacts of Family Rewards change during the next two years, after the incentives are no longer available.

- **Implementation and process analysis.** This analysis will explore the operations of the CCT program, focusing particularly on the roles and experiences of the implementing institutions (Seedco, City agencies, and community partners), and on the perceptions and experiences of the participating families. It will describe how the program was structured, how the partnerships that Seedco established with the Neighborhood Partnership Organizations functioned, how families were recruited to the program, how the incentives payment system was marketed and operationalized, and how families viewed and responded to the incentives offer. As part of assessing family responses, it will estimate incentive “take-up rates” across the education, health, and workforce areas, and seek to understand what factors promoted or impeded taking up the incentives. The study will also explore public perceptions of this program and how these views evolve over time, particularly among community leaders, policymakers, teachers, and other institutional actors
- **Benefit-cost analysis.** This analysis will estimate the cost of operating Family Rewards, distinguishing how much was spent on various aspects of program delivery versus the amount of cash transferred to the participating families. It will also make a number of benefit-cost comparisons, examining the economic “gains” and “losses” from several perspectives, such as from the perspective of participants and their families, and from the perspective of taxpayers and government budgets. It will look at benefits and costs that are directly observable during the period of data collection, as well as over a longer time horizon (e.g., 10 years), based on systematic projections using alternative assumptions about trends in costs and impacts.

Types of data

Data for this comprehensive analysis will come from a variety of sources and involve both quantitative and qualitative information. These data will include:

- **Background Information Form.** At the point of random assignment, data on all sample members on such matters as their Social Security identification number,

contact information, demographic background (for instance, marital status, ethnicity, citizenship status, education, and English proficiency), employment status and characteristics, and household income and composition were collected. The baseline data provide a means of assessing whether the and control groups, on average, represent similar types of people according to measurable characteristics, and for statistically controlling for anticipated small random disparities in characteristics that emerge between the groups. The data will also be used to understand the composition of the sample in each community, as well as to define key subgroups of sample members to determine whether the program ends up working particularly well for certain categories of low-income households, and whether receipt of the program's incentives varies in systematic ways across particular types of families.

- **Administrative records data.** Important data will be collected for all program and control group members from: (a) school records on attendance, grades, and test-scores from the NYC Department of Education; (b) New York State Unemployment Insurance (UI) quarterly wage records; (c) TANF, Food Stamps, and UI benefits records; and (d) Medicaid eligibility records. Other administrative data sources may be used as well, if it is deemed useful and feasible to do so. The study will collect administrative data for all sample members dating back several years prior to their date of random assignment into the study. The possibility of that a sizeable number of families in the program and control group will move out of New York City during the course of the demonstration (and that the incentives strategies may affect these choices) will make it important to explore the possibility of matching to state as well as City education and welfare records in addition to state UI data.
- **Client surveys.** The demonstration will hire a nationally reputed survey firm to conduct follow-up surveys of a randomly selected subsample of program and control group members. Resources permitting, the evaluation will include three waves of surveys: at 12 months after random assignment (to help address short-term incentives use and early program impacts); at about 36-months, which is at the end of the three-year program; and at about 60 months after enrollment, allowing for post-incentive follow-up on key outcomes of interest. For each wave, interviews will be administered through a mix of telephone calls and in-person follow-up, with the latter mode of interview reserved for sample members who are unavailable by telephone. An overall response rate of at least 80 percent will be sought for each survey wave. Interviews will be sought with families who move out of New York City to the extent practical.

Each survey wave will yield completed interviews with approximately 3,000 respondents (1,500 in the program group and 1,500 in the control group). These respondents will be allocated evenly across the three targeted grade levels. This means that, for each of the three targeted grade levels, interviews would be completed with 1,000 respondents—500 from the program group and 500 from the control group.

- The follow-up surveys will take approximately 40 minutes to complete, and they will be designed to capture information that is not obtainable through administrative records or any other tracking system developed for the project. They will cover a number of programmatic and substantive issues, clustered into several topical areas. Some will focus on questions just for the program group, such as perceptions of the incentives and the CCT program; receipt of the various types of incentives among different family members; impediments to receiving various incentives. Others will focus on service use and outcomes more generally, for both the program and control groups. For example, these would include children's school experiences and parental involvement in their education; certain child outcomes; family health care practices and health conditions; parental participation in skills training and educational programs; parents' employment, job quality, and earnings; and family income, poverty, and material well-being.⁵⁴ Where possible, the survey instruments will use well-tested and validated measures of key constructs.
- **Field observations and focus groups.** These will involve in-depth qualitative interviews with various stakeholder groups. Program staff (from Seedco and other institutional partners) will be interviewed to learn about operational strategies, challenges, and lessons about implementing the CCT program. Participants will be interviewed to learn about their perspectives on the program and experiences in it, including why they have or have not taken advantage of the services and incentives offered to them. External stakeholders, such as community leaders, policymakers, and others, will be interviewed to gauge their reactions to the program.

The Impact Analysis

The impact analysis will determine the effects of the intervention—on the overall sample and for key subgroups of families and children.

Key questions

The impact analysis will be structured to answer a number of key questions on the effectiveness of Family Rewards. These include the following:

- **Does the CCT increase children's performance in school?** The most direct effects of the CCT are expected for the outcomes that are incentivized. The impact analysis will examine the program's effects on regular attendance in school and behavioral problems. Effects on test scores might also arise directly through the incentive payments for performance or indirectly through higher attendance and increased parental homework monitoring and support for the student. They might also arise from higher family income achieved by the combined incentives package. Intermediate outcomes of interest will include time spent on homework and the use of tutoring services, at least as reported on the participant survey. Other positive

⁵⁴ For the most part, the design assumes that a parent will respond to the survey. If resources permit, MDRC is willing to consider the feasibility of adding a supplement that will be completed by a sample of children in one or all of the targeted grades.

effects might include reductions in special education referrals, reductions in grade retention, and (in high school) increased credits-for-age and reduced dropout rates.

- **Does the CCT increase parents' involvement in their children's education?** The key direct effect of the CCT should occur for parents' attendance at parent-teacher conferences. However, the impact analysis will examine more broadly parents' involvement in their children's schooling, including help with and monitoring of homework, for example (as measured through self-reports on the client survey, recognizing that there might be some reporting bias that would need to be addressed). Effects on these outcomes might stem from the incentives for better attendance and performance in school, but they also might arise through feedback effects if the children become more engaged in school and perform better over time.
- **Do the effects of the CCT vary by children's grade level?** Although the impact analysis will estimate various school performance and outcomes the pooled sample of children (regardless of grade level), a major objective of the analysis is to understand the program's effects on children at particular grade levels. In particular, the analysis will give priority to estimating the program's impacts on the "index children," distinguishing the effects for children who being Family Rewards in the fourth, seventh, or ninth grade. (The previously designed sampling plan was constructed with this in mind.) Moreover, it will consider whether the effects of the program grow, stabilize, or decline as those children advance to subsequent grades.

Because siblings will also be offered the educational incentives, the impact analysis may be able to estimate effects for children at other individual grade levels, sample sizes permitting. At a minimum, we can estimate effects for three age *ranges*—elementary, middle school, and high school. These three ranges are very distinct in terms of the development stages they represent and challenges faced by low-income children. Prior research also showed that the effects of incentives programs on children varied by the children's age. The results of tests for differences in impacts across these groups will indicate whether the effectiveness of the program varies by school level.

Finally, assuming a three year intervention, using the overlapping sample of ninth graders (those who entered the evaluation as ninth graders and those who entered as seventh graders) we can compare impacts for these two cohorts to assess the relative effects on ninth grade performance of *beginning* the intervention earlier than ninth grade versus beginning in ninth grade. Sibling sample size permitting, we may be able to perform this type of analysis for other grade levels.

- **Does the CCT increase access to and use of health care?** A key goal of the CCT is to encourage families to find and maintain health insurance, with the incentives for private insurance somewhat greater than those for Medicaid. The impact analysis will examine the CCT's effects on families' enrollment in and use of both

private and public insurance. It will also measure effects on families' visits to the doctor for emergency and preventive care, either at hospitals or at non-emergency providers.

- **Does the CCT improve health outcomes for adults and children?** If the CCT encourages take up and sustained enrollment in health insurance and the use of preventive care visits, the next question is whether this translates into improved health outcomes for both adults and children. The impact analysis will examine the program's effects on a variety of survey-based health measures, including overall health rating, vaccination status of children, and the presence of certain chronic conditions. (The option of obtaining administering records on some health care outcomes is also being explored.)
- **Does the CCT increase adults' employment, earnings, and participation in education and training activities?** Results from earlier incentives evaluations suggest that supplements conditioned on work can increase employment, employment stability, and earnings. The impact analysis will examine the CCT's effects on overall employment rates, full-time employment, sustained employment, and job stability. The incentives for full-time, stable might also affect job quality, as measured by wages and benefits, for example. The CCT's effects on employment and earnings might also arise indirectly through better family health outcomes and even through children's improved performance in school. We will also examine effects on participating in education or training activities while working. Note that the training incentives might also lead to an increase in part-time work.
- **Does the CCT increase family income, reduce poverty and decrease material hardship?** The idea behind the full package of incentives is to increase income both in the short- and long-term. The impact analysis will examine the program's effects on total income, income sources, and indices of material hardship like food insecurity.
- **Does the CCT improve educational and health outcomes for pre-school aged children?** Many siblings of the targeted children who were enrolled in the study were in the 0-5 age group at the time of random assignment (over 1,500 children in total in the program and control groups). It may be possible to determine whether Family Rewards has any positive effects on young children. Such effects might emerge if, for example, the program increases the likelihood of preventive health care, which, for young children, may mean a greater likelihood of receiving age-appropriate developmental screening and appropriate referrals for early intervention. The program might also have positive impacts on young children by improving family income. Over time, these effects might help increase the school-readiness of young children as they enter elementary school.
- **Do the effects of the CCT vary by family type, neighborhood, or other key characteristics?** We will estimate the program's impacts on subgroups of

families defined by key baseline characteristics. The age of the children, for example, may affect not only children's response to the education incentives, but also parents' responses to the employment incentives. In addition, the effects of previous financial incentives programs were found to differ between boys and girls, suggesting that the child's gender may matter. Other important characteristics include race/ethnicity, work history, and area of residence. An additional subgroup to consider is school quality: do effects on children's education outcomes vary by the quality of the school they attend, as measured, for example, by average test scores and attendance at those schools? This analysis will depend on having adequate sample sizes and variation in quality across schools.

- **What are the pathways through which the CCT achieves its effects?** In addition to assessing the direct impact of the program on key outcomes, we will consider examining impacts on certain ultimate outcomes (such as school performance) via impacts on more intermediate outcomes (such as health status). Although we will explore various non-experimental methods for tracing these kinds of relationships, we will likely need to rely on a more descriptive assessment using variation in effects across geographic areas or subgroups.⁵⁵ For example, we might examine whether those geographic areas that had larger effects on health status also had larger effects on education outcomes.

Research design and analysis issues

The power of the experimental design for the CCT evaluation comes from the fact that, when implemented properly, random assignment ensures that the treatment and control groups are alike in all aspects of the distribution of observed and unobserved baseline and pre-baseline characteristics. As a result, any post-baseline differences between the two groups can be interpreted as effects of the intervention.

- **Analytical approach**

Our proposed analytical approach combines the power of a random assignment design with statistical modeling to improve the efficiency of the estimates and to address additional questions. The impact analysis will focus on differences in the outcomes of treatment and control group members.

- **Basic impacts.** The first tool for understanding the effects of the intervention is to compare average outcomes for the program and control groups. This is a straightforward calculation that is generally easy to explain to policymakers and other nontechnical audiences.

⁵⁵ The method of instrumental variables analysis (used by MDRC for a series of welfare-to-work demonstrations to investigate the causal pathways that link child well-being to intermediate outcomes such as parents' employment, welfare receipt, and income) is unlikely to be feasible for this study, given the limited number of areas in which the program is being tested.

- **Conditional outcomes.** Some outcomes can be measured for only a subset of the sample. For example, hourly wage rates can be measured only for workers. For such outcomes, the experimental framework can be preserved by analyzing distributions. For example, we could compare whether people who worked and earned above a certain amount represented a higher proportion of the program group than of the control group (rather than comparing average hourly wages among workers in both groups, who may not represent the equivalent types of people). Other conditional outcomes can include attendance for students still in school and type of health care facility visited, among those who made some visits.
- **More sophisticated methods.** Most random assignment studies attempt to increase the precision of estimated basic impacts by adjusting for baseline characteristics. In estimating these effects, we can use a linear regression framework, or a more complex set of methods depending on the nature of the dependent variable and the type of issues being addressed, such as: logistic regressions for binary outcomes (e.g., whether or not someone works); Poisson regressions for outcomes that take on only a few values (e.g., months of employment); quantile regressions to examine the distribution of outcomes for continuous outcomes, such as test scores and earnings, and income; and duration (hazard) models for outcomes that depend on an event, such as the conditional probability of entering employment over several periods.⁵⁶

Our evaluation plan will also include a variety of methods to test the robustness of the findings, including:

- **Understanding the efficacy of the intervention.** Comparing outcomes for all randomly assigned families provides an estimate of the effectiveness of the intervention in real-world settings in which some families do not take advantage of the incentives. It might also be important to understand the effects on those who take up one or more of the incentives.⁵⁷ We will explore various nonexperimental methods to estimate the effect of the program on those who received payments, including nonparticipation adjustments,⁵⁸ instrumental variable techniques,⁵⁹ and propensity score matching.⁶⁰
- **Multiple measures.** The evaluation will examine many outcomes across a number of domains. When multiple outcomes are examined, the probability of finding statistically significant effects increases, even when the intervention has no effect. For example, if 10 outcomes are examined in a study of an ineffective treatment, it is likely that one of them will be statistically significant at the ten percent level by chance. Although the statistical community has not reached consensus on the

⁵⁶ In notation, the basic impacts are calculated from a regression of the form $y_i = \alpha + \beta E_i + \delta X_i + \varepsilon_i$ where y_i is the outcome for individual i , E_i equals one for those assigned to the treatment group and 0 otherwise, and X_i is a set of baseline characteristics. The parameter β measures the effect of the treatment.

⁵⁷ These concepts are also known as the impact of the intent to treat (ITT) and the impact of treatment on the treated (TOT).

⁵⁸ Bloom, 1984; Peck, 2003.

⁵⁹ Greene, 1999; Duncan et al., 1998; Angrist, Imbens, and Rubin, 1996.

⁶⁰ Rosenbaum and Rubin, 1983; Dehejia and Wahba, 1999; Michalopoulos, Bloom, and Hill, 2004.

appropriate method of correcting for this problem, we will review the literature on this issue and consider making such corrections using an appropriate method.⁶¹

- **Subgroup comparisons.** The evaluation will investigate whether the intervention worked especially well for particular subgroups of families or areas. MDRC conducts subgroup analyses in most of its random assignment studies. Subgroup impacts can be calculated in three ways. In “split-sample” subgroup analyses, the full sample is divided into two or more mutually exclusive and exhaustive groups, such as families with young children at the point of random assignment versus those with older children. Impacts are estimated for each group separately. In addition to determining whether the intervention had statistically significant effects for each subgroup, Q-statistics will be used to determine whether impacts differ significantly across subgroups (Hedges and Olkin, 1985). A related type of subgroup analysis uses regression methods to see if the effects of the intervention vary significantly with a continuous baseline measure (or one that takes on many values), such as age or initial attendance levels. Finally, “conditional” subgroup analyses take this idea one step further by controlling for the effect of other baseline characteristics when estimating the relationship between a particular subgroup and program effects. For example, in estimating whether the program has larger effects for families with young children, it might be important to control for the program’s varying effects based on parents’ age, education, and work history.⁶²
 - **Pattern of impacts over time.** The evaluation will collect data for as long as five years after random assignment, and it will be useful to understand how the effects of the policy change over time. For example, the program’s effects might fade or grow over time, and the pattern of change, on educational impacts, might differ by the child’s grade level at baseline. In addition, it will be of great interest to know whether any effects of the CCT persist beyond the families being eligible for incentives.
- **Achieving adequate statistical power**

A well-designed evaluation should include adequate sample size to detect policy-relevant impacts. The sample sizes available for this study should be adequate for detecting effects that are relatively small or modest (but meaningful from a policy standpoint) on a range of outcomes, both for the full sample as well as for subgroups by grade level and by key demographic characteristics.⁶³

⁶¹ A useful overview of the issue is provided by Darlington (1990).

⁶² In notation, the basic impacts are calculated from a regression of the form $y_i = \alpha + \beta E_i + \delta X_i + \varepsilon_i$ where y_i is the outcome for individual i , E_i equals one for those assigned to the treatment group and 0 otherwise, and X_i is a set of baseline characteristics. The parameter β measures the effect of the treatment. For subgroup analysis with a continuous subgroup measure, the regression would take the form $y_i = \alpha + \beta E_i + \gamma Z_i E_i + \delta X_i + \varepsilon_i$. Here, γ indicates how impacts vary with the baseline characteristic, and Z_i is a particular baseline characteristic for which subgroup impacts are being estimated. Conditional subgroup analysis can be represented by the equation $y_i = \alpha + \beta E_i + \gamma Z_i E_i + \delta X_i + \varepsilon_i$.

⁶³ Detailed calculations of minimum detectable effects are included in an earlier version of this paper and are available upon request.

Key outcomes and their measurement

Figure 1 provided a list of key domains that may be affected by the CCT. Table 8 presents a list of outcomes and their data sources specific to the impact analysis. The data sources include follow-up surveys of evaluation enrollees at 12 and 24 months after random assignment⁶⁴; records data from the New York City public school system; records data from the UI and welfare systems; and Medicaid eligibility records data. Records data will be collected for at least five years following each sample member's date of random assignment. The surveys will provide data on outcomes that will not be available from other sources, such as parents' reports of children's school progress and their engagement in their children's school performance, children's engagement in school and educational aspirations, health outcomes, and job characteristics. They will also be a secondary source of data on some outcomes, a useful redundancy given the likelihood of gaps in other data sources.

Within each broad domain, the outcomes are ordered according to whether they are considered intermediate or final. The first and most basic question that the impact analysis will answer is: "Were treatment group members more likely than those in the control group to reach the key milestones and receive incentive payments?" (Analysis of incentive receipt will be discussed in the section describing the implementation study.)

External validity and a non-participant study

Although the random assignment design of the program evaluation provides the most credible evidence on its effectiveness, only a fraction of eligible families will be enrolled. The experiment thus provides an unbiased estimate of the impacts of the program on families who attended orientation sessions and agreed to participate in the evaluation, but not necessarily on all eligible families. Depending on resources and access to appropriate data, the evaluation will attempt to examine the external validity of the study and assess whether the results can be generalized to the full eligible population, a critical issue should the CCT be rolled out on a broader scale. This analysis would require access to Department of Education data on non-volunteers from the sampling frame. (To avoid transgressing privacy rules, we will explore whether the analysis could be done in collaboration with the Department's own researchers.)

The first step in this process is to assess the incidence of non-participation. Since the study was not designed to serve all eligible families, the rate of non-participation will be considered within the context of sample size constraints. The second step is to assess how selective the participation process is, or the extent to which non-participants differ from participants. For this analysis, we can use the full set of available observable characteristics, both at baseline (comparing non-participants to the combined sample of treatment and control

⁶⁴ Pending results from the first two survey waves, we will consider fielding a third wave of the survey to capture longer-term effects.

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Table 8

Examples of Outcomes for the Impact Analysis (Selected List)

Outcome	Data Source
Education	
Attendance and behavior in school	School records data
Parent's attendance at parent-teacher conferences	Follow-up surveys
Time spent on homework and receipt of tutoring services	Follow-up surveys
Test scores, grades, promotion, and graduation	School records data
Engagement in school and educational aspirations	Follow-up surveys
Post-secondary education - children	Follow-up surveys
Post-secondary education - adults	Follow-up surveys
Health	
Obtaining and maintaining health insurance	State records data; Follow-up surveys
Preventive care visits	Follow-up surveys
Health	Follow-up surveys
Workforce	
Employment stability and hours worked	UI records data; Follow-up surveys
Participation in education and training	Follow-up surveys
Earnings, wage rates, job quality	UI records data; Follow-up surveys
Employment and earnings - youth	Follow-up surveys
Other	
Family income and well-being	Follow-up surveys
Material hardship	Follow-up surveys
Parental depression	Follow-up surveys
Family structure	Follow-up surveys
Residential mobility (e.g., into lower-poverty neighborhoods)	Follow-up surveys

families) and post-baseline (comparing non-participants to the control families only). Key variables of interest, which will come from administrative records, will include pre and post-baseline labor market outcomes and children's school performance.

It is hoped that marketing and outreach will be extensive enough to bring in a relatively wide range of families. Nonetheless, there is always the possibility that participating families will differ significantly from non-participants. In this case, although we would know, for example, that the program is effective for those who volunteered for it as a pilot project, we would be limited in our ability to predict its effectiveness for all eligible families—more of whom might volunteer if the program were instituted as an ongoing public policy. Should the analysis reveal notable differences between the two groups, we would seek to examine in more depth, through the implementation research, the ways in which participants differ from non-participants and the reasons for non-participation. Is non-participation largely an information issue, with some families less likely to know about the program? Do rates of non-participation vary across neighborhoods or are they associated with particular marketing strategies? Which types of families are less likely to participate? Knowledge about who participates and who does not will inform efforts to implement similar programs more broadly.

The Implementation and Participation Analysis

The implementation research will address several major topics critical to the success of Family Rewards. These lines of inquiry will help document what the intervention provides, assess the fidelity of program operations to the intervention design, support the interpretation of impact findings, and provide the data to measure program participation and project costs. The topics include:

- Recruiting families and starting up Family Rewards
- Operating the CCT verification and payment system
- How participants experience and respond to Family Rewards
- How families use Family Rewards resources
- How policymakers, community leaders, and other stakeholders view Family Rewards over time

Table 9 presents the combination of qualitative and quantitative data that will be used to examine each of the major themes of the implementation and participation analysis. As shown, data will be collected through in-depth interviews, program observations, focus groups, Seedco's participant tracking and verification database, and structured client surveys. Various types of interviews will be conducted over the course of the demonstration with Seedco and Neighborhood Partner Organization staff as well as with representatives from various stakeholder groups such as teachers, community leaders, and local policy-makers. Interviews and possibly focus groups will also be conducted of program participants, (and of non-participants if low levels of participation in the program become a problem), to determine how and why people do or do not take advantage of the services and resources that Family Rewards offers. These interviews will shed light on the features, conditions, and challenges of operating a conditional cash transfer program, including program staffing and management requirements, operating practices, and institutional relationships, as well as perceptions of the program among

participants and the wider community. These are all factors that can influence the program's ability to deliver a cash transfer program efficiently and effectively. Observations of program services and staff/customer interactions and examinations of program records (for instance, Seedco's monitoring and verification system) will shed light on the program flow and engagement and participation in Family Rewards.

Recruiting families for Family Rewards

- *How is the concept of a conditional cash transfer marketed to eligible families and what encourages or discourages them from volunteering from the program? How are marketing strategies adapted to suit cultural and other differences across the targeted communities, and how are these appeals received?*

Recruitment and enrollment in Family Rewards, as with any initiative, is a critical component to successful implementation. The lead Neighborhood Partner Organizations in each community district conducted the recruitment drive (collaborating with other neighborhood organizations and Seedco), and implemented the enrollment and random assignment process. Within Family Rewards, recruitment and enrollment dynamics may be especially critical, as the “web” of community organizations charged with recruitment will also become an information and referral network for participants. Since part of the initiative's theory of change is that participation will make families more likely to seek help, training, or other services, the robustness of these networks and their ability to deliver value-added information to participants is very important.

Our research will thus examine the ways that recruitment rolls out in the various neighborhoods, the ways that the community networks form, and how participants come to know about the program, and what appealed to them to apply for it—or to decline the invitation.

Family Rewards

Table 9

Data Collection Methods for the Implementation and Participation Study

Key Issues	Qualitative Data			Quantitative Data	
	Seedco staff interviews	Focus groups	Interviews with NPOs and program partners	Client Survey	CCT Payment System
Recruitment and Start up of Family Rewards - Marketing and outreach - Call-in center, hotline, walk-in services, and program web site - Adequacy of staff tools and training - NPOs, service networks, and community resources	X	X	X	X	
Operating CCT Verification and Payment System - Challenges in implementing payment procedures / system - Strategies for engaging and monitoring participants - Customer attitudes	X	X	X	X	X
Participation / Use of Family Rewards Resources - Characteristics of participants - Type of services and supports most often utilized - Contacts/interactions with program staff - Strategies to promote participation	X	X	X	X	X
Other Stakeholders' Perspectives of Initiative		X			
Counterfactual Environment: - Supports and services available for control group - Participation level of control group in similar activities and supports	X	X	X	X	

Illustrative questions:

- What outreach strategies did the NPOs use to publicize and communicate about the program to the various groups of low-income households targeted for the program? How did they encourage them to apply? How were marketing efforts tailored to suit different neighborhoods and racial and ethnic groups?
- How did the NPOs deal with bad contact information on families? What strategies did they employ to locate hard-to-find families?
- Which approaches were most successful in getting families, once contacted, to come forward and enroll in the study? Can differences in outreach efforts account for any variations in enrollment patterns across communities?
- Do participants mention specific features of the intervention that encouraged them to apply?
- Why did some potentially eligible families *not* want to become involved with the program?
- How did various NPOs work with Seedco and other neighborhood organizations and institutions to form a recruitment network, and what was involved in creating or operating such a network?

Operating the CCT verification and payment system

- *How does the CCT verification and payment system function, and how does it advance (or impede) the goals of the program? How does it manage potential tensions between participants, the intermediary, and supporting institutions?*

By intent, Family Rewards is designed to side-step traditional case management institutions and to have an influence on families through its cash rewards system. At the same time, the creation of effective monitoring and verification systems for participant compliance with the conditions that earn rewards, and the development of efficient delivery mechanisms for the cash payments, are likely to be critical to the success of the endeavor. The quality of these interactions—between Seedco and participants, and between participants and those asked to provide documentation for Seedco⁶⁵—will certainly inform the experience of the program for participants.

Classic studies on learning processes emphasize that immediacy of reward is an important component to behavior change. Seedco will likely be challenged to create systems for verification and payment that tie desired actions to timely receipt of the income supplement.⁶⁶ In

⁶⁵For example, health care providers, school administrators, or employers.

⁶⁶ That is, within the 2-month window envisioned in program design and not lagging so that the payments occur much later than the desired action. To some extent, a longer than desired lag time will be unavoidable owing to delays in getting the necessary information to Seedco, either because of late coupon submission, or, in the case of

a related vein, the specific ways that monitoring and payment are implemented represent points of potential friction and dispute that must be managed systematically.⁶⁷ Participants' interactions with Seedco and the NPOs may also influence the program's success in improving participant outcomes. For example, if participants experience their participation in monitoring and verification to be a relatively smooth process by which they can gain additional income without excessive transaction costs, they may be more willing to pursue the activities and rewards the program is promoting. In contrast, negative experiences with the program and dashed expectations may discourage families to the point that participant outcomes are adversely affected, perhaps exacerbating what many studies on intrinsic versus extrinsic sources of motivation have shown to be challenges in providing reward incentives to bring about longer-term positive changes.⁶⁸

Also important is the sheer complexity of administering a claims process for a rewards system made up of a complex menu of incentives. Much of the documentation of meeting conditionalities come in the form of submitted paper coupons that must be carefully reviewed by Seedco staff. Rigorous quality control and data security procedures must also be established, and a process for responding to participants' queries and complaints must be put into place and operated well. Understanding Seedco's experiences in trying to build and operate such a system will be crucial for understanding the feasibility of replicating the program on a larger-scale, and how best to do so.

Illustrative questions:

- How does the system for monitoring and verification of behaviors work in practice? What major challenges were encountered and how were these addressed?
- How does the intermediary manage tensions with participants? How cumbersome is the process for participants and for the intermediary?
- How do relevant actors (health care providers, educators, and employers) view documentation obligations? Do they tend to provide documentation when asked? Is there a sense of stigma or dynamic of discrimination that comes into play?
- In general, how successfully are payments processed within the two-month window envisioned by program designers? When delays happen, what are barriers to timely payment?

school records data, delays in processing those data at the Department of education. For example, months will pass after students take their standardized tests before these tests are scored and the information transferred to Seedco.

⁶⁷ To understand how conflict-laden some of these interactions can be, one need only recall the experience of verification and the delivery of specific welfare grants before the advent of more general cash support in the 1960s (Piven and Cloward 1971).

⁶⁸ Benabou and Tirole 2003.

Patterns of incentive receipt and use of other assistance

- *To what extent do participating families receive the available cash transfers, how does receipt vary by CCT component, and what factors influence the degree of receipt?*

This aspect of the implementation research will assess, from a quantitative perspective, how much participants are actually receiving or taking advantage of the core elements of the CCT intervention.

Illustrative questions:

- How do the incentive “take-up rates” vary over the education, health, and workforce areas? What conditions and milestones are participants most likely and least likely to achieve?
- Do many families maximize their incentives? In other words, do they meet all the conditions to draw down the maximum possible incentive that they could earn?
- How does the degree of receipt of incentives vary over time—for example, does it grow or decline between the first and third years of the program?
- Does the nature and degree of participation vary by family type, neighborhood, or other key characteristics? If so, in what ways?

How participants experience and respond to Family Rewards

- *How do families experience participation in Family Rewards? How do they see their lives changing or remaining the same because of Family Rewards?*

At the most basic level, research will address the experience of participants and around the Family Rewards program—examining what it means to families to take part in it, how they view the incentives and resultant behavior changes, how they feel about various program components, and the ways that they see their lives changing because of participation in Family Rewards. Related to these broad questions, research will also seek to understand how the program may be influencing parent-child relationships and the dynamics between families and school officials, given that money that is offered to the family depends on the children’s efforts and their performance in school.

This research will attempt to shed light on important underlying dynamics that might help explain the program’s broad impacts (or the lack of effects). For example, within the literature on work incentives, there are instances in which programs are unable to induce participants to take up a subsidy, for various reasons including participants’ desire to maintain more of a work-home balance. In cases where there is little take-up of the incentive subsidy, qualitative research may

seek to understand whether participants feel it is *possible* to advance their health, education, and workforce goals, but also what barriers exist to greater participation in the initiative. For example, the Family Rewards theory of change suggests that parents who are faced with incentives around schooling will seek after-school programs, workshops, homework help, or other types of services to help improve school performance. It will be important to learn how well the program communicates information on these opportunities to participants (through its “information and communications” component)—including whether the “messages” from the program are “heard” and understood by participants—and how much families try to take advantage of services that might help them meet the program’s conditions for incentives payments.

The implementation analysis will thus examine whether Family Rewards payment amounts and conditionalities are the right ones to appeal to poor families, whether the conditionalities are attainable, and whether they are even understood by those to whom they are directed, and what participants do to try to maximize the amounts they earn. In this regard, it is also very important to determine why some program participants choose *not*—or are unable—to take advantage of the incentives the program offers. (Both survey and qualitative data will be used to explore these issues.)

Another important and potentially controversial assumption of Family Rewards is how it seeks to change “habits” through cash incentive, and how “habits” formed during the incentive period will endure after the period ends. For example, around such issues as school attendance, Qualitative research will examine the extent to which participants view the links between defined behavior changes and the monetary benefit; whether it ‘cheapens’ or makes more desirable the activity; and how participants see their lives and actions after the incentive period is set to end.

Illustrative questions:

- What are participants’ primary perceptions of Family Rewards and the critical elements of the initiative? How well do they understand the incentives offer and conditions, and how they can get assistance in meeting those conditions?
- How do participants view the operation and burdens of the verification system, and the efficiency of the payments system?
- How accessible are the various community services that correspond to the types of incentives that Family Rewards offers, and how helpful do participants feel those services are for improving their family’s education, health care, or workforce outcomes?
- How does Family Rewards play out in areas that may be especially sensitive to intervention (internal family dynamics, parent-school dynamics, or dynamics with employers)? Are there any aspects of the program that seem to be resulting in unintended negative consequences in these areas?

- How enduring do participants feel ‘habits’ will be after the incentive period ends, and in which areas do they think changes will be most enduring or effective?

How families use Family Rewards resources

- *What does the cash incentive “buy” participants, and how does this help explain the success of the program?*

Especially for families who are successful in achieving program goals, the cash supplement can represent a significant enhancement of family income. The presence of this income supplement itself is an important variable that may help support participants efforts to do the very things the program encourages. For example, the incentive could be used for increasing the amount or quality of child care, for buying tutoring services, paying health insurance co-payments, paying for transportation, or for buying clothing suitable for work, all of which may help families make progress in the areas of schooling, health care, and work.

Research will attempt to assess how much participants tend to use their cash transfers in these ways, and to also learn what else the additional income “buys” them—in terms of housing stability, food security, or other assistance—that may support progress toward other desired program impacts. For example, a family that does not have to move repeatedly or that has less stress related to living in poor quality housing may be able to focus on health maintenance, educational improvement, or career advancement. To the extent that the income supplement provides significant additional resources to families, we will also seek to understand how families experience greater well-being as a result of this income support.

Illustrative questions:

- How do participants use the income supplement? How does the use of the supplement advance the goals of the program?
- Do participants view greater income security as helping them concentrate on the program’s desired activities?
- In what areas do participants see the income support as making a difference in their lives?
- What are measurable qualitative areas of quality-of-life improvement, around housing stability, freedom from debt, or other general forms of other types of stability?

How policymakers, community leaders, and other stakeholders view Family Rewards over time

- *What are the perceptions of Family Rewards by key partners and other stakeholders, including policymakers, community groups, and other community leaders?*

In addition to understanding how participants see their participation in the initiative, we hope to explore the extent to which the program's goals and strategies align with or diverge from those of various community institutions and organizations, city agencies, and community and political leaders. This is important for assessing the potential for replicating the program as a broader City policy if the evaluation finds it to be effective.

Illustrative questions:

- Which aspects of Family Rewards are most supported by key stakeholders? Which are most controversial?
- Among key potential stakeholders, are there any issues around community support that should be considered with regard to the potential replication of the initiative?

The Benefit-Cost Study

The decision to implement Family Rewards as an actual program on a broader scale will depend on whether the benefits of the intervention are deemed to justify the costs. The evaluation plan thus includes a comprehensive benefit-cost analysis.

This section introduces the key elements of the proposed benefit-cost analysis, briefly describes different perspectives traditionally used to evaluate benefits and costs, and discusses the data requirements and some challenges inherent in assessing the transfer cost efficiency of a program like Family Rewards.

Gross and net costs and benefits

The cost analysis will estimate the average *gross* and *net* costs of Family Rewards per program participant. Gross cost refers to the value of resources expended to operate the program. The incentive payments made to participants will account for the lion's share of these operating costs of Family Rewards (assuming high take-up rates). The resources required to administer the process of delivering those payments from collecting and processing documentation, verifying compliance, authorizing payments, making the fund transfers, and so on, will also be key cost. In addition, managing the flow of participants through the program (beginning with outreach and recruitment), operating the program's information and communication component, and addressing customer service problems, are all key cost components. The study will thus estimate these and other costs, which will be incurred to varying degrees by Seedco, the Neighborhood Partner Organizations, key financial partners, and by certain City agencies (e.g., the Department of Education, for supplying data to verify compliance with the educational conditionalities).

Family Rewards offers no case management or other kinds of direct service interventions. However, it is expected that the program's incentives will increase participants' use of various kinds of services that may be publicly funded or subsidized. To take one example, the program will encourage participants to take up training or education courses. The resources involved in providing these and other services must be accounted for, even though they are not directly an

operating cost of Family Rewards. Thus, total gross costs attributable to Family Rewards will be made up of the combination of program operating costs and other related service costs.

Net costs refer to the expenditures on program participants that are above and beyond expenditures for alternative services and supports that would have been made on behalf of program participants in the absence of the program. Thus, estimating net costs requires estimating the gross costs estimated incurred by the control group—which will primarily be the costs of their use of relevant services used and subtracting that amount from the total gross costs incurred by the program group.

The gross costs of the program will be useful to administrators and policy analysts interested in understanding the overall budget for Family Rewards expenditures, which will be essential in any planning for replicating the program. However, judgments about whether the benefits of the demonstration “justify its costs” will depend upon the estimated net costs of Family Rewards per program participant, and how these compare to the program’s net “benefits.”

In benefit-cost analysis, the “benefits” side of the equation usually refers to the monetary value of the average effects, or impacts, of the program on critical outcome measures. Although some impact estimates (e.g., impacts on earnings, welfare payments, Food Stamp payments) will supply the core monetary values needed for the computation of benefits, the analysis will impute the value of impacts on other economic outcomes that are not directly measured but that can be inferred using data from a combination of sources (e.g., tax payments and EITC amounts). In some cases, as discussed below, it will not be possible to monetarize the effects of the program (e.g., any impacts on measures of social and psychological well-being). However, any such effects must still be considered in weighing the overall merits of the program.

Three perspectives on the benefits and costs of the program

A given program effect may be interpreted as a gain for one segment of society and a loss (or a neutral outcome) for another. Thus, benefit-cost analysis requires assigning the costs and benefits of a social program to the appropriate constituency. Three distinct benefit-cost perspectives are key:

The *Program Participant Perspective* identifies the anticipated net gains and losses for program group members, indicating how they fared as a result of the program. In the case of Family Rewards, participants’ CCT payments, net earnings increases, and net increases in EITC payments would represent gains for participants, whereas higher net tax payments (resulting from earnings gains) and reductions in welfare payments would represent losses. A program may be considered as have produced an overall net economic gain from the standpoint of program group members if the total economic gains exceed the losses. In essence, the benefit-cost analysis will aim to determine whether the program participants “come out ahead” economically because of Family Rewards.

The *Government and Taxpayers’ Perspective* typically identifies the anticipated gains and losses for the taxpayers, who ultimately pay for such programs and for the institutions

responsible for administering the program. Although the operating costs of Family Rewards will be largely borne by a consortium of private funders, they can be viewed as proxies for the costs the government would be incurring if the program were operating as a government policy. With this in mind, the total net cost of Family Rewards (as described above) would represent a net loss to the government. Extra government expenditures for subsidized health insurance or any other benefits (not already accounted for on the net cost side of the ledger) will also be an economic loss from the government's perspective. At the same time, any public savings from reductions in certain benefit receipt (e.g., TANF), or in the costs of emergency room use and hospitalizations, or reductions in juvenile arrests and detention, or in reductions in reliance on special education, and so on, would represent gains from the government budget or taxpayer perspective, as would any increase in tax payments by participants. The sum of these gains and losses will indicate whether the savings offset the losses and, hence, whether the government budget "comes out ahead" as a result of the program.

In assessing the return on the government's projected expenditure on Family Rewards, it is important to consider how the results vary across different levels of governments—i.e., New York City, New York State, and the federal government—which are financially responsible in different ways for various services, transfers, public safety, and public health, tax receipts, and tax credits. Thus, the benefit-cost analysis will aim to assess the separately the return on investments separately for each of these levels of government.

A third major perspective is that of society as a whole, which is usually estimated by summing the net gains or losses across the participant and taxpayer groups. This perspective ignores the distributional effects of the program and focuses more on its overall economic efficiency—that is, whether the economic gains for both groups exceed the economic losses for "society" as a whole.

Non-economic benefits and cost-effectiveness

If all of the program's benefits and costs could be measured in dollars, the benefit-cost analysis could produce a neat "bottom line" of the program's overall accomplishments. However, Family Rewards endeavors to affect family and child well-being—for example by improving participants' health, improving children's academic achievement, and, ultimately, improving family's overall quality of life. Applying a dollar value to such non-economic and quality of life outcomes is challenging and controversial. Rather than placing a monetary value on these types of effects, the evaluation will discuss the non-financial impacts of the program along side the economic gains and losses, and in relation to the program's cost. The goal here will be to help policymakers see the "full picture"—the program's economic and non-economic effects—so that they can make their own informed judgments about the overall merits of the program.

It is also important to note that some anti-poverty programs such as Family Rewards deliberately seek to transfer income to its target population (for example, in the form of cash incentives and increased receipt of Food Stamps and the EITC) to help alleviate hardship. They do not necessarily aim to save the government or taxpayers money, at least in the short term, or are viewed as having an inherent value (e.g., public education, public safety, and public health).

In such cases, the net “loss” of resources to the government may, instead be viewed more properly as an “investment,” but not necessarily one that is intended to pay for itself on a dollar-for-dollar basis. Thus, as part of the benefit-cost analysis, a set of estimates will be calculated to show how impacts relate to those investments. For example, this analysis would answer such questions as, “what is the average percentage reduction in poverty per net dollar invested; what is the average percentage reduction in poor health outcomes per net dollar invested, and what is the average increase in a measure of child achievement or family well-being per dollar invested?”

The benefit-cost and cost-effectiveness assessments of Family Rewards will take into account the possibility that the program will be more successful for some participant subgroups than for others. To the extent that data and resources permit, the costs and benefits may be analyzed and presented separately for certain subgroups—for example, for the separate grade-level cohorts, or for families who entered the program while on TANF versus those not on TANF, and so on.

Analysis plan for the benefit-cost analysis

The benefit-cost analysis builds upon the implementation and impact analyses, and it uses data from a variety of sources, including the administrative records used for parts of the impact analysis, participant survey data (especially to gauge the use of services among program and control group measures, as well as for measuring non-economic impacts), and various types of fiscal data from Seedco, the Neighborhood Partner Organizations, and from the health, education, and welfare systems.

The results from the analysis will be presented to cover specified time horizons—in other words, specific periods of time after participants enroll in the program. It is important not to set this time horizon too short, because most program costs are likely to be incurred in the shorter term, while the full program benefits may take much longer to emerge. At the same time, it may be that certain positive program impacts begin to fade, or “decay,” after a few years, if, for example, control group members “catch up” to program group members on key outcomes over time. For this study, the overall benefit-cost results will probably be estimated to cover five and ten years after random assignment, and perhaps longer. As with any forecasting, the precision of longer-term estimates will depend on the variability of the behavior and the length of time for which it has been observed, and that precision diminishes progressively the further into the future the projections are made. Estimates must also be adjusted for inflation and discounted to convert them to present values. A number of sensitivity tests will help establish how sensitive the results are to different assumptions in the projection formulas.

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